Fine-Grained Text and Spectral Attribute Supervised Semantic Network for Hyperspectral Image Zero-Shot Domain Generalization

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***Abstract*—Cross-scene hyperspectral image (HSI) classification has attracted increasing attention due to domain shifts caused by distribution differences and label scarcity. However, most domain generalization (DG) methods assume that classes are shared across domains, which fails when novel categories appear. Zero-shot DG (ZSDG) is thus essential to address both semantic and distribution shifts. Recent advances in vision-language foundation models (VLFMs) offer promising solutions to this challenge. In this article, we propose a ZSDG method, FTSA, which efficiently fine-tunes the CLIP model via prompt learning (PL) to extract a domain-invariant semantic space from textual and spectral modalities. Specifically, we introduce two interclass shared learnable prompts in the text branch. To guide PL of shared contextual features, we use large language model (LLM)-derived land-cover priors and intrinsic spectral attributes as dual complementary supervision signals. By designing several chain-of-thought (CoT) templates, the LLM automatically generates well-decoupled, fine-grained land-cover descriptions. To compensate for the weak representation of highly nonlinear spectral correlations in text, we design the spectral attributes local cache (SALC) module, construct a fixed attribute cache, and introduce an attribute update strategy based on local sparse matching to dynamically aggregate generic spectral visual attributes. Parallel prompts and category names are concatenated to align the corresponding textual and spectral features during training, implicitly learning universal spatial–spectral semantics. On multiple datasets, FTSA achieves superior performance compared to state-of-the-art methods, demonstrating robustness and zero-shot generalization. The code will be released as open source at https://github.com/blackdyc/FTSA**

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***Index Terms*—Hyperspectral images (HSIs), prompt learning (PL), vision-language foundation models (VLFMs), zero-shot domain generalization (ZSDG).**

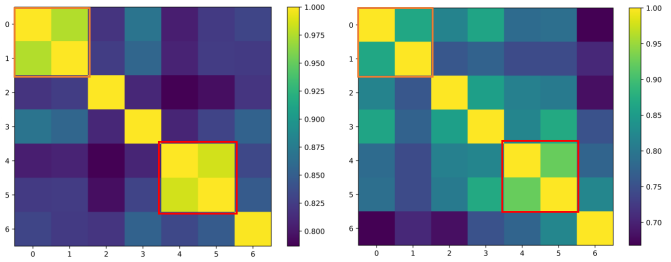
I. Introduction

HYPERSPECTRAL images (HSIs) are remote sensing data with high spectral resolution, typically comprising hundreds of bands that capture rich spatial–spectral information. HSIs have been widely applied in environmental monitoring [1], agriculture [2], urban planning [3], and mineral exploration [4].

Despite the success of deep learning in closed-set HSI classification (HSIC), domain-specific shifts caused by sensor variation, weather, and geography result in performance drops in unseen domains. Moreover, the high labeling cost and limited availability of HSI data hinder generalization to target domains, thereby restricting large-scale deployment. Domain generalization (DG) has shown great promise for cross-scene HSIC by training only on labeled source data and deploying models directly to target domains. Depending on the number of source domains, DG is categorized into single-source (SSDG) and multisource (MSDG) settings. In this article, we focus on the more challenging SSDG scenario.

The core of DG is to improve model generalization by learning domain-invariant features from source data. Most prior work [5], [6], [7], [8], [9] focuses on extracting robust domain-invariant representations from images using data augmentation and style randomization. However, unimodal vision-based methods often suffer from unstable performance under scene and category shifts, as hyperspectral visual features are sensitive to illumination, environment, and resolution, which introduces biases that hinder generalization. Inspired by advances in vision-language foundation models (VLFMs), recent studies have explored using text supervision to enhance domain-invariant feature learning. Zhang et al. [10] perform cross-scene HSIC using coarse-grained templates and manually crafted fine-grained land-cover descriptions. Wang et al. [11] build local and global text templates to capture multilevel semantics. Wang et al. [12] combine fixed templates with visually guided fine-grained semantics for dynamic scene adaptation. However, most existing DG methods that leverage the textual modality to learn shared semantic embeddings for cross-scene HSIC still face several limitations.

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(a) (b)

Fig. 1. Heatmap of semantic feature similarity among land-cover classes on the Houston dataset. (a) Heatmap obtained using fixed text templates. (b) Heatmap obtained using fine-grained description generated by the LLM. Notably, pairs such as "grass healthy" versus "grass stressed" (marked by orange boxes) and "residential buildings" versus "nonresidential buildings" (marked by red boxes) exhibit high similarity, posing classification challenges. Compared to fixed text templates, fine-grained semantic features generated by the LLM demonstrate clearer interclass distinctions.

Limitations in Constructing Text Descriptions: Text descriptions are usually built from fixed templates like "an HSI of {class}" or manually crafted fine-grained features for each land-cover class, resulting in suboptimal performance and high labor costs. To our knowledge, no prior work has used large language models (LLMs) to generate fine-grained textual descriptions for cross-scene HSIC. Leveraging LLM prior knowledge, a few prompt templates (e.g., "What does {class} look like in HSIs?") suffice to generate rich descriptions for each class, greatly reducing manual annotation costs. Moreover, compared to fixed templates, fine-grained descriptions offer stronger interclass separability, particularly for visually similar classes. Fig. 1 presents heatmaps of feature similarity for each class on the Houston dataset, comparing descriptions generated by fixed text templates [Fig. 1(a)] and LLMs [Fig. 1(b)] after CLIP [13] text encoding. As shown in the figure, category pairs such as "grass healthy" versus "grass stressed" (orange boxes) and "residential buildings" versus "nonresidential buildings" (red boxes) in the Houston dataset are challenging to distinguish due to their shared superclasses and strong visual correlation. Semantic features from fixed templates show weak discriminability for these pairs, as they contain only basic class names with limited semantic information. In contrast, fine-grained semantic features generated by LLMs exhibit clear differentiation.

Limitations in Generalizing to Novel Categories: Most existing studies [10], [11] employ the textual modality primarily as an auxiliary constraint, using the text embedding space as a domain-invariant semantic space to guide the image encoder in learning domain-invariant features. However, they rely on explicit classifiers such as linear or 1-D convolutional layers for final classification, which weakens the inherent open-vocabulary capability of vision-language dual-tower models and restricts generalization to unseen categories.

However, in real-world scenarios, novel classes frequently emerge in the target domain, requiring the model to generalize from seen classes to unseen ones across domains. Therefore, in this article, we explore a further SSDG setting, zero-shot SSDG (ZS-SSDG), which has been rarely investigated in the current hyperspectral community. ZS-SSDG requires the model to access only the labeled samples of the seen classes in the source domain during training and then transfer directly to the unseen classes of the target domain. We refer to the task corresponding to the ZS-SSDG setting as zero-shot cross-scene HSIC, which will be defined in detail in Section II-A.

To address this task, we propose a fine-grained text and spectral attribute supervised semantic network, FTSA. This is a parameter-efficient fine-tuning (PEFT) method based on prompt learning (PL), which efficiently adapts the pretrained CLIP model to HSI distributions and enables zero-shot generalization via image–text alignment. FTSA introduces prompt vectors into both the image and text branches to enable efficient adaptation. On the image side, prompt vectors are adaptively generated by a spectral prompt generation (SPG) module based on spectral features. On the text side, two learnable prompt vectors replace the fixed templates used in prior work. To prevent overfitting of text-branch prompts to source-domain image distributions, we introduce domain-invariant, category-level fine-grained textual supervision using LLM-based prior knowledge. In addition, we design a spectral attributes local cache (SALC) module for storing generic, generalizable spectral attributes. Spectral tokens from the image encoder, carrying fine-grained semantics, are used to momentum-update attribute entries in the cache via a local sparse matching strategy during training. The learnable parallel prompts separately align fine-grained descriptions and spectral attributes, which are then fused into a domain-invariant semantic space. Visual features are projected into this space to align with textual features, thereby enabling zero-shot cross-scene HSIC. The main contributions of FTSA are summarized as follows.

1) We propose FTSA, a zero-shot DG (ZSDG) method that introduces two parallel learnable soft prompts in the text branch to align domain-invariant representations from textual and spectral modalities. This enables the learning of a robust shared semantic space and achieves zero-shot cross-scene HSIC via image–text matching.
2) To guide the soft prompts to learn generalizable features, we leverage external knowledge from LLM to generate domain-invariant, fine-grained textual descriptions of land covers. To improve decoupling and diversity, we design several chain-of-thought (CoT) templates to guide LLM generation, enabling the automatic production of high-quality textual supervision for training.
3) To address the limited representation of highly nonlinear spectral correlations in text, we design the SPG and SALC modules. The SPG module leverages spectral information via similar-band clustering, reduces high-dimensional redundancy, and generates spectral tokens using 1-D convolution. The SALC module builds a fixed attribute cache and employs a local sparse matching-based update strategy to dynamically aggregate similar spectral tokens, capturing universal spectral visual attributes.
4) Our method follows the PL paradigm and achieves state-of-the-art performance on three benchmark datasets for zero-shot cross-scene HSIC, using only 0.99 M–1.23 M

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trainable parameters (approximately 1.3% of the total model parameters).

The structure of this article is organized as follows. Section II reviews the related work on cross-scene HSIC, VLFMs, and PL. Section III presents the proposed method in detail. Sections IV and V report comprehensive experimental results. Finally, Section VI concludes this article and outlines directions for future work.

II. Related Work

A. Cross-Scene HSIC

1) Closed-Set Cross-Scene HSIC: To mitigate the performance drop of traditional single-scene HSIC methods under distribution shifts between training and testing data, the cross-scene HSIC task has been proposed. Depending on access to target domain data, cross-scene HSIC methods fall into domain adaptation (DA) and DG. Given the high cost of acquiring HSI data, this study focuses on DG-based approaches, which assume no access to target domain samples during training. Such methods aim at extracting domain-invariant representations of features from training samples of the source domain. Zhang et al. [5] extract domain-invariant features using generative adversarial learning and supervised contrast learning. Xu et al. [14] propose a novel dynamic modulation model to solve the imbalanced convergence in multimodal data classification, which contributes to the fusion classification of remote sensing data. Gao et al. [9] propose an invariant semantic DG mashup network that learns robust domain-invariant semantics through feature style covariance and spatial stochastic discriminants in combination with a two-sampling direct adversarial contrast learning strategy.

Recent advances in vision-language methods offer new solutions for cross-scene HSIC. LDGnet [10] enhances performance by aligning images with both coarse- and fine-grained textual descriptions. However, generating fine-grained texts requires manual feature design, which is time-consuming and difficult to scale. EHSnet [11] builds an explicit semantic network by designing local and global text templates, aligning image features with high-level semantics to improve DG in cross-scene HSIC. However, the global template assumes identical class distributions across domains, thereby limiting generalization to novel classes. TASA-Net [12] learns domain-invariant semantics by combining handcrafted templates with visually guided fine-grained prompts, using gated weighting to balance semantic granularity in a shared embedding space. However, evidence from natural image domains suggests that fixed templates often introduce irrelevant features, which can impair cross-scene HSIC performance.

2) Zero-Shot Cross-Scene HSIC: As previously noted, this study focuses on the ZS-SSDG setting, referred to as zero-shot cross-scene HSIC in the hyperspectral domain. While related to the cross-domain zero-shot (CDZS) HSIC setting explored in prior work [15], [16], key differences exist. In CDZS HSIC, the target domain includes both relevant and irrelevant tasks, with annotated samples from the irrelevant tasks available during training. In contrast, the problem considered here assumes no access to any target domain data and requires much stronger generalization. We will provide a formal definition of the zero-shot cross-scene HSIC problem in Section III-A.

B. Vision-Language Foundation Models

In cross-scene HSIC, traditional methods typically extract visual features via CNN or ViT models. However, visual features are sensitive to variations in imaging environment, sensors, and other external factors, complicating the separation of category-related features from background and noise [17]. Consequently, the textual modality provides more robust semantic representations and superior feature disentanglement, being less influenced by scene variations and offering complementary attribute features to visual data.

In recent years, VLFMs have become a focal point in multimodal learning by pretraining on large-scale image–text pairs (e.g., 400 M pairs) to learn joint cross-modal representations, demonstrating strong transfer capabilities in downstream tasks. Mainstream VLFMs fall into single- and dual-stream architectures. Single-stream models (e.g., VisualBERT [18] and UNIMO [19]) embed images and texts into a shared semantic space using a unified Transformer for joint encoding. Dual-stream models (e.g., CLIP [13] and ALIGN [20]) employ separate image and text encoders, aligning modalities via contrastive learning. For instance, CLIP constructs positive and negative sample pairs in the projection space, and optimizes representations to bring positives closer while pushing negatives apart. Compared to image-only supervised models, VLFMs leverage linguistic supervision to achieve cross-modal semantic alignment, significantly enriching visual representations.

Recently, several VLFMs have been proposed in remote sensing [21], [22]. Although these models demonstrate strong representation learning capabilities, their adaptation to the HSI domain remains challenging. Most existing VLFMs are tailored for natural images and struggle to efficiently capture the spatial–spectral joint features of HSI. Moreover, due to the complexity and fragmented feature relationships in HSI, constructing effective textual modality representations remains a critical problem. In cross-scene HSIC, recent vision-language multimodal models [10], [12] have advanced joint representation learning, yet their visual pipelines remain suboptimal. Because of inherent structural differences between hyperspectral and natural images, these methods still depend on retraining specialized HSI feature extractors. This approach increases model complexity and substantially enlarges the parameter space, resulting in significantly higher training costs. Considering that pretrained VLFMs (e.g., CLIP) have been trained on large-scale natural image–text pairs, they possess strong general-purpose feature extraction capabilities. Therefore, when adapting HSI for downstream tasks, we can directly utilize this feature through the PEFT paradigm. For downstream tasks of HSI, we only need to adapt the frozen pretrained model by injecting the ability to understand high-dimensional spectral information, without requiring full fine-tuning or model redesign. This efficient fine-tuning not only reduces the training cost and design complexity, but also enables effective adaptation to HSI-related tasks.

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C. Prompt Learning

PL [23], [24], [25], [26], [27], [28], [29] is an efficient strategy for adapting VLFMs by replacing manual hard prompts with learnable soft prompts, allowing rapid adaptation of models like CLIP via fine-tuning a small set of parameters. The pioneering CoOp method [23] substitutes handcrafted category descriptions (e.g., "a photograph of a {class}") with learnable textual prompts, enabling efficient few-shot classification fine-tuning. Subsequent works propose various learnable prompts to fine-tune text [30], image [31], or joint image–text features [32] in VLFM outputs for domain-specific adaptation.

Adapted VLFMs often suffer from degraded zero-shot generalization due to prompt overfitting on seen class distributions. To balance recognition accuracy between base and novel classes, recent studies introduce constraints and external knowledge to enhance generalization. For instance, CoCoOp [24] employs visual feature-conditioned prompts that generate dynamic, instance-specific constraints to better adapt to unseen classes. Similarly, ArGue [33] uses an LLM to produce detailed class descriptions, enriching the text encoder with fine-grained semantic information.

Most PL methods have been explored in the natural image domain, while strategies tailored to the unique characteristics of HSI data remain limited. In zero-shot cross-scene HSIC, the primary challenges lie in the spectral nonlinear shifts across domains and the presence of unseen target categories. Although previous studies have demonstrated that LLMs can introduce fine-grained spatial features such as position, shape, texture, and edges through their powerful prior knowledge, guiding LLMs to generate well-decoupled descriptive texts remains a major challenge for data modalities like HSI, which contain both spatial and spectral complexity. Furthermore, our preliminary experiments suggest that LLMs are often limited in representing the nonlinear spectral dependencies inherent in HSI. Given that the image modality inherently captures rich spectral information, another key challenge lies in constructing an effective supervision mechanism that enables the model to learn universal and generalizable spectral representations.

III. Proposed Method

In this section, we first provide a formal definition of the zero-shot cross-scene HSIC problem. We then present an overview of the overall architecture of our proposed FTSA. Next, we describe each stage of the model in detail. Finally, we discuss the optimization objectives during training and the inference process for achieving zero-shot classification in the target domain.

A. Problem Description

Assume that the source domain (D_s) contains land-cover categories $C_s = [C_1; C_2, \ldots, C_{N_s}]$, and the target domain (D_t) contains land-cover categories $C_t = [C_1; C_2, \ldots, C_{N_t}]$, where N_s and N_t are the number of categories in the source and target domains, respectively, and $C_s \cap C_t = \emptyset$. D_s comprises numerous labeled samples. During training, a certain proportion of labeled samples $X_s = \{(x_i, y_i)\}_{i=1}^{N_s \times N} \subseteq D_s$ are randomly selected from each category of D_s, where N is the number of training samples per category. In the inference stage, the trained model is directly applied to the novel-class samples $X_t = \{(x_i, y_i)\}_{i=1}^{M_t} \subseteq D_t$, where M_t is the total number of samples in D_t.

B. Overall Framework

The overall flowchart of the proposed FTSA model is shown in Fig. 2. FTSA integrates spatial and spectral pathways to learn complementary domain-invariant representations. In the *spatial–spectral feature extraction* stage, the source-domain HSI data is segmented into $13 \times 13 \times$ c patches (c denotes the number of spectral bands). RGB bands are extracted and resized to 224×224 using bilinear interpolation to meet the CLIP input requirements. Simultaneously, the SPG module first generates compact spectral tokens by reducing redundancy. These spectral tokens are concatenated with the spatial tokens from CLIP's patch embedding and jointly fed into the frozen CLIP image encoder for spatial–spectral feature extraction. The image encoder subsequently outputs refined spectral tokens, which are mapped into the textual modality through a modality alignment module. This aligned representation is further used to update the spectral attribute entries maintained by the SALC module, whose cache is continuously refined through local sparse matching and momentum-based aggregation.

In the *fine-grained text generation* and *domain-invariant text representation learning* stages, LLM-generated textual descriptions provide supervision for the spatial pathway. These fine-grained texts capture domain-invariant spatial characteristics of the base classes, and when encoded by the frozen CLIP text encoder, they yield stable domain-invariant spatial semantic features for each category. Meanwhile, the spectral attributes cached in the SALC module offer complementary supervision for the spectral pathway. Two interclass-shared learnable soft prompts are concatenated with category names and fed into the frozen CLIP text encoder—one prompt primarily aligns with spatial semantics, and the other aligns with spectral attributes. Through joint optimization, these prompts implicitly learn class-agnostic and generalizable spatial–spectral representations. Finally, in the *image–text matching* stage, the textual features produced by the two prompts are fused via element-wise summation to form a unified semantic representation, which is then matched with the output of the image encoder to perform land-cover classification.

C. Spatial–Spectral Feature Extraction

Let the HSI be denoted as a 3-D cube $\boldsymbol{X} \in \mathbb{R}^{H \times W \times C}$. Each sampled patch $\widetilde{\boldsymbol{X}} \in \mathbb{R}^{p \times p \times C}$, where p and C denote spatial resolution and spectral channels, respectively, is incompatible with CLIP, which is pretrained on $224 \times 224 \times 3$ natural images. To adapt, we resize the patch to 224×224 via bilinear interpolation and extract the RGB bands to match CLIP's input format.

While HSI's spectral vectors encode rich land-cover attributes, discarding non-RGB bands weakens spectral semantic modeling. To compensate, the SPG module constructs a

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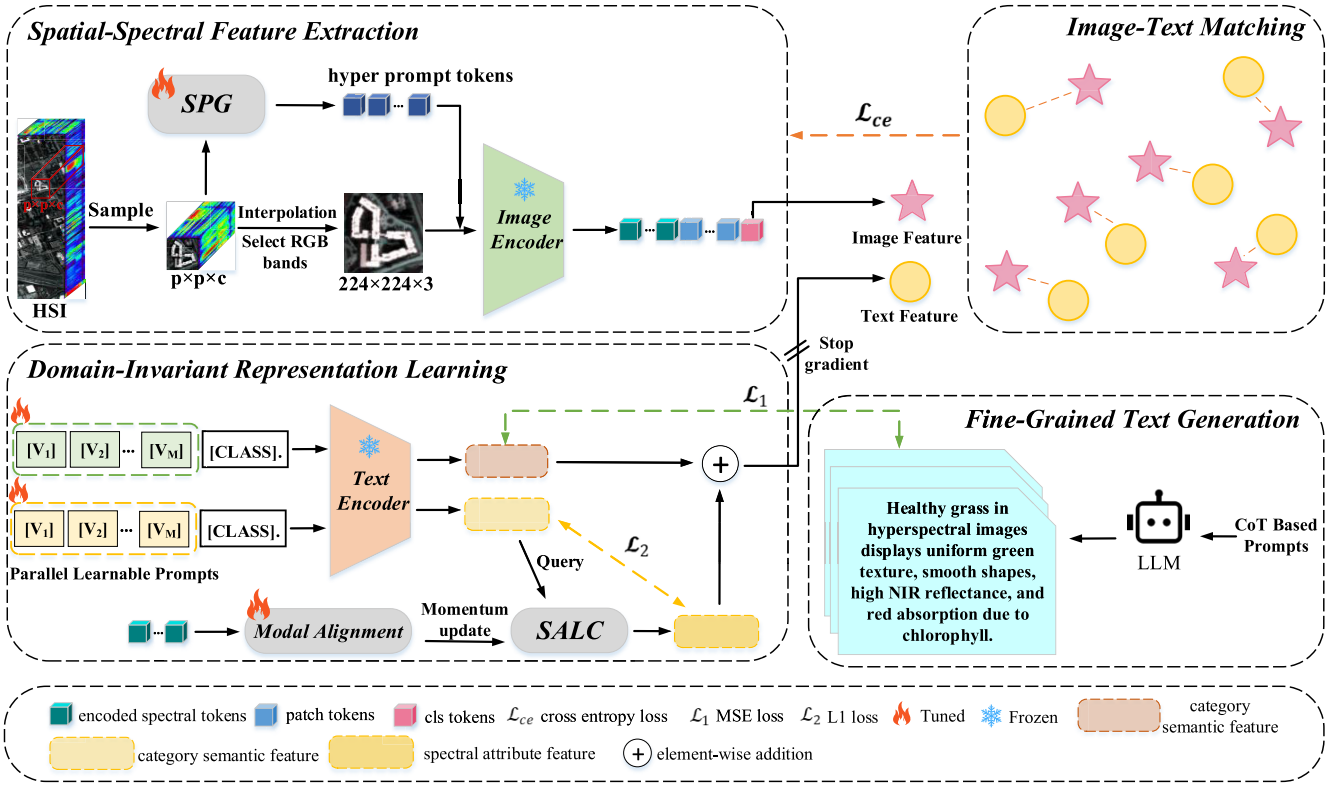


Fig. 2. Flowchart of FTSA, which mainly consists of four stages. The SPG module in spatial–spectral feature extraction dynamically constructs bands as prompt tokens. The question templates based on CoT designed for fine-grained text description generation guide the LLM to generate well-decoupled text descriptions of land covers. The SALC module designed for domain-invariant semantic representation learning aggregates spectral tokens gradually into corresponding attribute items to store spectral visual attributes through an update strategy based on local sparse matching. Two parallel learnable soft prompts are introduced to align fine-grained texts and spectral attributes in cache, whose outputs are fused to form a shared semantic space for image–text alignment.

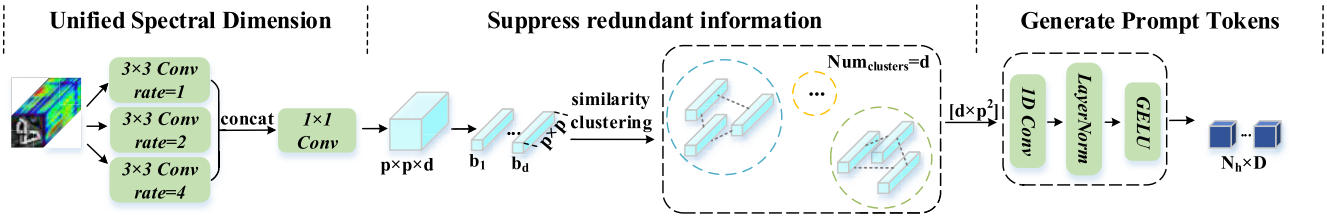


Fig. 3. Diagram of the SPG module.

spectral prompt $\boldsymbol{H} \in \mathbb{R}^{n\times D}$ by grouping high-dimensional bands, where n is the number of spectral tokens. This prompt is concatenated with CLIP's patch embedding $\boldsymbol{Z} = [\boldsymbol{z}_1; \boldsymbol{z}_2, \ldots, \boldsymbol{z}_N] \in \mathbb{R}^{N\times D}$ and passed to the encoder for feature extraction

$$\boldsymbol{E}^l = g^l\left(\left[\boldsymbol{Z}^{l-1}, \boldsymbol{H}^{l-1}\right]\right), \quad l = 1, 2, \ldots, L \tag{1}$$

where $N = 224/P \cdot 224/P$, $g^l(\cdot)$ denotes the lth layer transformer, $L = 12$, and $P = 32$ in ViT-B-32 version.

As shown in Fig. 3, the SPG module addresses sensor-dependent band variations by employing parallel multiscale feature extraction and fusion layers to project spectral inputs to a unified dimension c (e.g., 100). It applies three parallel 3×3 dilated convolutions with dilation rates $d \in 1, 2, 4$ to extract multiscale spectral features $\boldsymbol{F}_1$–$\boldsymbol{F}_3$ from the HSI patch $\widetilde{\boldsymbol{X}}$. These features are concatenated along the channel dimension and fused via a 1×1 convolution to produce $\boldsymbol{F}_{\text{fused}} \in \mathbb{R}^{p\times p\times c}$.

To address the redundancy and uneven distribution of informative content across hyperspectral bands, we apply a dynamic k-NN graph [34] to cluster spectrally similar bands and enhance salient features while suppressing irrelevant ones. Each band is compared with others using cosine similarity, and its top-k neighbors are selected to form a local band cluster. Within each cluster, nonlinear transformations extract edge features between the center band and its neighbors, which are then aggregated via symmetric max pooling to retain the most discriminative information. This results in a compact representation that emphasizes meaningful spectral cues. Finally, a large-kernel 1-D convolution, followed by layer normalization and activation, aggregates features along the spectral dimension and maps them to the CLIP embedding

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space, thereby producing spectral prompt vectors for subsequent spatial–spectral feature extraction.

D. Domain-Invariant Semantic Representation Learning

1) Preliminaries: In PL methods such as CoOp [23], manually crafted hard prompts are replaced with learnable soft prompts, formulated as

$$\boldsymbol{P} = [\boldsymbol{V}_1][\boldsymbol{V}_2], \ldots, [\boldsymbol{V}_M][\mathrm{CLS}] \tag{2}$$

where $\boldsymbol{V}_1, \boldsymbol{V}_2, \ldots, \boldsymbol{V}_M$ are learnable tokens and CLS is the hard class token.

In addition to embedding learnable tokens at the input layer, they can also be introduced into deeper transformer layers. For the ith layer, the previous prompt tokens are replaced with the layer-specific learnable tokens $[\boldsymbol{V}_1^i, \boldsymbol{V}_2^i, \ldots, \boldsymbol{V}_m^i]$, yielding the prompt vector $\boldsymbol{P}^i$ for that layer

$$\boldsymbol{P}^i = \left[\boldsymbol{V}_1^i\right]\left[\boldsymbol{V}_2^i\right], \ldots, \left[\boldsymbol{V}_M^i\right][\mathrm{CLS}_{i-1}]. \tag{3}$$

For learnable soft prompts, appropriate supervision is crucial to progressively capture universal and generalizable implicit features during gradient-based optimization.

2) Fine-Grained Domain-Invariant Text Generation: Due to the significant domain gap between HSI and natural images, it is necessary to design specialized prompt templates tailored to HSI characteristics to guide the LLM in generating feature descriptions (here, prompts refer to those used to elicit text from LLMs, distinct from prompts in PL). Although this step requires some manual effort, only a few generic templates are needed—unlike prior work such as [10], which manually crafted multiple fine-grained descriptions for each land-cover class.

The same land cover may exhibit heterogeneous features across scenes due to varying imaging conditions and geographic contexts. However, certain attributes that strongly correlate with land-cover types remain stable across domains, known as domain-invariant features. For instance, while rice fields may differ in appearance under varying lighting conditions or growth stages, their high reflectance in the red-edge and near-infrared bands remains consistent. To accurately capture domain-invariant features, LLMs should be guided to focus on stable spatial attributes (e.g., shape, texture, edges, and color) and spectral characteristics (e.g., reflectance patterns, spectral curves, absorption peaks) that remain consistent across diverse environments. Therefore, we design several prompt templates based on the CoT [35] to guide the LLM in deeply analyzing the multilevel domain-invariant features of each category. The two designed templates are shown in the following.

Template 1: What does {class} look like in HSIs? Please first analyze the scenarios and locations where this type of land-cover commonly appears, then analyze its spatial characteristics such as shape, texture, and color in HSIs, and then analyze its spectral characteristics. Finally, conclude with a single-sentence summary of your analysis using no more than 80 tokens.

Template 2: Describe a {class} as it would appear in an HSI. Please first analyze the scenarios and locations where this type of land-cover commonly appears, then analyze its spatial characteristics such as shape, texture, and color in HSIs, and then analyze its spectral characteristics. Finally, conclude with a single-sentence summary of your analysis using no more than 80 tokens.

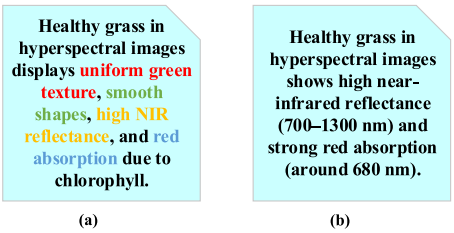


Fig. 4. LLM-generated fine-grained descriptions of "healthy grass" (CoT prompts versus regular prompts). LLM guided by CoT prompts shows better decoupling in multilevel feature descriptions. (a) Generation based on CoT prompts. (b) Generation based on ordinary prompts.

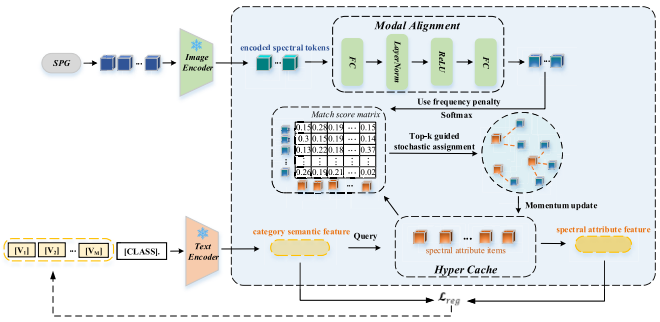


Fig. 5. Diagram of the SALC module.

We adopt a staged cognitive guidance strategy to decompose the complex HSI terrain description task into a progressive analysis process of "scene localization to spatial features to spectral features," which can overcome the problem of information mixing or omission in traditional prompt engineering (as shown in Fig. 4). This strategy enhances the decoupling of fine-grained descriptions and facilitates the learning of generalized representations through soft prompts. In this work, we employ the DeepSeek–V3 [36] model as our LLM; however, any pretrained LLM can be effectively utilized for this purpose. A generation temperature of 0.8 is used to encourage greater diversity in the generated category descriptions.

3) Spectral Attribute Feature Injection: The complex nonlinear spectral characteristics in high-dimensional data are often challenging to describe verbally. However, the spectral prompt tokens $\boldsymbol{H}$, introduced in Section III-D, capture these features via the image encoder. Mapping $\boldsymbol{H}$ to a learnable prompt complements the limitations of LLM in spectral description and enhances the adaptability of the text embedding space to spectral shifts and unseen categories. Therefore, we design the SALC module, as shown in Fig. 5. We maintain a local cache in which K spectral attribute items $\boldsymbol{S} \in \mathbb{R}^{K \times D}$ are stored. These attribute items are initially randomly initialized and then L2-normalized. During training, the cached attribute items are updated by $\boldsymbol{H}$ output by the image encoder so that each item can learn and represent different spectral characteristics.

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Algorithm 1 Spectral Attribute Cache Update

Require: Spectral prompt $\boldsymbol{H} \in \mathbb{R}^{n\times D}$, Spectral attribute cache $\boldsymbol{S} \in \mathbb{R}^{K\times D}$, usage counter $\boldsymbol{\gamma} \in \mathbb{R}^{K}$, penalty weight λ, momentum coefficient α

Ensure: Updated spectral cache $\boldsymbol{S}$

Map spectral prompt $\boldsymbol{H}$ to text modality: $\widetilde{\boldsymbol{H}} \leftarrow$ Modality Alignment($\boldsymbol{H}$)

$\boldsymbol{S}' \leftarrow$ clone($\boldsymbol{S}$)

Normalize

Input: $\hat{\boldsymbol{H}} \leftarrow$ Normalize($\widetilde{\boldsymbol{H}}$)

Compute similarity: $\mathbf{Sim} \leftarrow \hat{\boldsymbol{H}} \cdot (\boldsymbol{S}')^{\top}$

Adjust similarity with penalty: $\mathbf{Sim}' \leftarrow \mathbf{Sim} - \lambda \cdot \boldsymbol{\gamma}$

Compute weights: $\mathbf{W} \leftarrow$ Softmax($\mathbf{Sim}'$)

for $i = 1$ **to** n **do**

  Select top-k indices and weights from $\mathbf{W}_i$

  Randomly pick one index from top-k

  Build sparse weights $\widetilde{\mathbf{W}}$ keeping only chosen indices

end for

Normalize rows of $\widetilde{\mathbf{W}}$

for $j = 1$ **to** cache size K **do**

  if $\sum_i \widetilde{\mathbf{W}}[i, j] > 0$ **then**

  Weighted average:

$$\boldsymbol{v} \leftarrow \frac{\sum_i \widetilde{\mathbf{W}}[i, j]\hat{\boldsymbol{H}}_i}{\sum_i \widetilde{\mathbf{W}}[i, j] + \epsilon}$$

  Update cache entry:

$$\boldsymbol{S}'_j \leftarrow \alpha \cdot \boldsymbol{S}_j + (1 - \alpha) \cdot \boldsymbol{v}$$

  Update usage counter:

$$\boldsymbol{\gamma}_j \leftarrow \boldsymbol{\gamma}_j + \sum_i \widetilde{\mathbf{W}}[i, j]$$

  end if

end for

Normalize updated cache: $\boldsymbol{S} \leftarrow$ Normalize($\boldsymbol{S}'$)

The update process begins with mapping the spectral tokens $\boldsymbol{H}$ to the text modality via a modality alignment module. This module includes two fully connected (FC) layers, a normalization layer, and an activation function. After modality alignment, the spectral semantic feature $\widetilde{\boldsymbol{H}}$ is obtained and represents a refined spectral feature in the text space

$$\widetilde{\boldsymbol{H}} = \mathrm{FC}\left(\mathrm{ReLU}\left(\mathrm{Norm}\left(\mathrm{FC}\left(\boldsymbol{H}\right)\right)\right)\right). \tag{4}$$

To achieve dynamic modeling of the spectral attribute space, we design an attribute update strategy based on local sparse matching, and its algorithm is described in Algorithm 1. First, the similarity score between each spectral token $\widetilde{\boldsymbol{h}}_j \in \widetilde{\boldsymbol{H}}$ and each attribute item $\boldsymbol{s}_i \in \boldsymbol{S}$ is calculated using the cosine similarity metric

$$\mathrm{sim}\left(\widetilde{\boldsymbol{h}}_j, \boldsymbol{s}_i\right) = \frac{\widetilde{\boldsymbol{h}}_j \cdot \boldsymbol{s}_i}{\left\|\widetilde{\boldsymbol{h}}_j\right\| \|\boldsymbol{s}_i\|}. \tag{5}$$

To avoid the collapse of spectral attribute space caused by frequent selection of certain attribute items, we introduce a dynamic frequency penalty term $\boldsymbol{\gamma} \in \mathbb{R}^{K}$, which is multiplied by a preset penalty coefficient λ (e.g., 0.05) and applied to the similarity

$$\mathrm{sim}\left(\widetilde{\boldsymbol{h}}_j, \boldsymbol{s}_i\right)' = \mathrm{sim}\left(\widetilde{\boldsymbol{h}}_j, \boldsymbol{s}_i\right) - \lambda \cdot \boldsymbol{\gamma}. \tag{6}$$

Then, the adjusted similarity matrix is subjected to a softmax operation to obtain the matching weights between spectral tokens and attribute items. Based on these matching weights, we select the top-k attribute items with the highest responses for each spectral token and randomly sample one of them as the assignment result of the spectral prompt. Through this assignment, each prompt token $\widetilde{\boldsymbol{h}}_j$ is assigned to an attribute item for updating. Let the attribute index assigned to the bth token be i, with matching weight w_b, constructing the sparse weight matrix $\widetilde{\mathbf{W}}$

$$\widetilde{\mathbf{W}}_{bj} = \begin{cases} w_b, & j = i \\ 0, & \text{otherwise} \end{cases}, b = 1, \ldots, n, j = 1, \ldots, K. \tag{7}$$

The update of the attribute items in the cache is carried out using a momentum-based update rule. Specifically, for each attribute item $\boldsymbol{S}_i \in \boldsymbol{S}$ in the cache, the following update is applied:

$$\boldsymbol{S}_i \leftarrow \alpha \cdot \boldsymbol{S}_i + (1 - \alpha) \cdot \left(\frac{\sum_b \widetilde{\mathbf{W}}_{bi} \cdot \widetilde{\boldsymbol{h}}_b}{\sum_b \widetilde{\mathbf{W}}_{bi} + \epsilon}\right) \quad \forall i \text{ with } \sum_b \widetilde{\mathbf{W}}_{bi} > 0 \tag{8}$$

where α is the momentum coefficient, ϵ is a small constant to prevent division by zero, and the term $\widetilde{\mathbf{W}}_{bi}$ represents the weight assigned to each spectral token based on its match with attribute item i. This momentum update allows each cached attribute item to be gradually refined over time based on the spectral features it is matched with.

Finally, the frequency of each attribute item's usage is updated dynamically, providing the basis for adjusting the penalty term $\boldsymbol{\gamma}$

$$\boldsymbol{\gamma}_i \leftarrow \boldsymbol{\gamma}_i + \sum_b \widetilde{\mathbf{W}}_{bi}. \tag{9}$$

This dynamic update mechanism allows the spectral cache to evolve over time, gradually aggregating spectral features with similar patterns, forming generalized spectral semantic attributes. This process improves the model's ability to generalize across domains and to new classes, as it enables the model to leverage shared spectral properties that are independent of specific categories.

The collaboration between SPG and SALC establishes a closed-loop synergy. SPG acts as a high-fidelity generator, where spectral tokens serve as discriminative queries to populate the SALC with representative spectral attributes. On the other hand, SALC functions as a spectral regularizer, ensuring that learnable text prompts maintain spectral consistency. This feedback loop anchors textual semantics in robust spectral evidence, preventing overfitting to spurious correlations and ensuring alignment with generalized spectral patterns.

To inject the knowledge from the spectral attribute cache into the text branch, we use the learnable prompt $\boldsymbol{P}_2$, which is concatenated with the category name $[\boldsymbol{P}_2, \mathrm{CLS}]$, to extract

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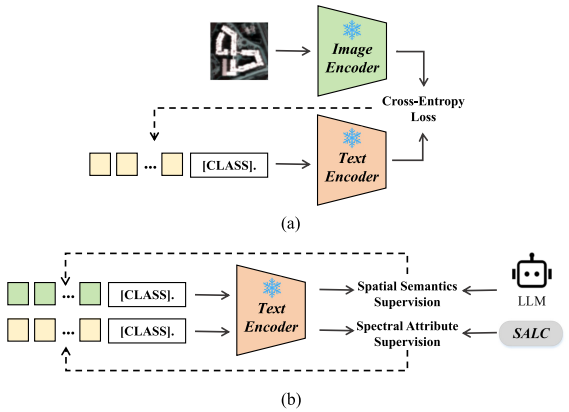


Fig. 6. Illustration of prompt optimization strategies. (a) Conventional PL methods and (b) our parallel dual-prompt optimization framework.

category-specific semantic features during forward propagation. The resulting feature $\boldsymbol{V}_2$ is treated as a query for a cross-attention operation with the spectral attribute items $\boldsymbol{S}$. This operation calculates the attention scores $\mathbf{A}$, which reflect the relevance between $\boldsymbol{V}_2$ and the cached spectral attributes

$$\mathbf{A}_{ij} = \frac{\exp\left(\boldsymbol{V}_i \cdot \boldsymbol{S}_j\right)}{\sum_{k=1}^{M} \exp\left(\boldsymbol{V}_i \cdot \boldsymbol{S}_k\right)} \quad \forall i \in [1, N], \quad j \in [1, M] \tag{10}$$

where N is the number of classes and M is the number of spectral attributes. These attention scores $\mathbf{A}$ are then used to compute a weighted sum of the spectral attribute items, resulting in the final spectral feature representation $\tilde{\boldsymbol{V}}$

$$\tilde{\boldsymbol{V}} = \mathbf{A} \cdot \boldsymbol{S}. \tag{11}$$

Although the attribute items in the cache are class-agnostic, the category name embedded in $[\boldsymbol{P}_2, \mathrm{CLS}]$ serves as semantic guidance for selecting the most relevant attributes from the spectral cache. This allows the learnable prompt $\boldsymbol{P}_2$ to focus on different subsets of the cache depending on the class semantics. As a result, the model can effectively retrieve the most suitable spectral characteristics for each class.

The representation $\tilde{\boldsymbol{V}}$ is element-wise summed with the output of the learnable prompt $\boldsymbol{P}_1$, introduced for aligning LLM-generated texts, to form the domain-invariant semantic representation $\boldsymbol{V}_F$.

E. Optimization Objectives

Existing prompt-based methods, such as CoOP [23] and CoCoOP [24], rely on task-specific losses like cross entropy to optimize prompts for image–text alignment, as illustrated in Fig. 6(a). However, this often leads to overfitting to the source domain, limiting generalization to unseen domains and classes. While approaches like PromptSRC [25] introduce constraints to balance base-class performance and generalization, they primarily focus on natural image tasks and overlook the unique properties of spectral data.

In contrast, our method avoids task-specific losses and instead introduces dual supervision signals from fine-grained textual descriptions and the SALC module, as shown in Fig. 6(b). These signals guide the alignment within a unified textual semantic space from complementary perspectives, where LLM descriptions emphasize spatial semantics and SALC captures spectral semantics, thereby improving generalization to unseen domains and new categories through the context mapping loss $\mathcal{L}_{\mathrm{map}}$ and spectral regularization loss $\mathcal{L}_{\mathrm{reg}}$.

On the text side, category names are concatenated with prompts as $[\boldsymbol{P}_1, \mathrm{CLS}]$ and $[\boldsymbol{P}_2, \mathrm{CLS}]$, and the CLIP text encoder outputs the corresponding semantic features $\boldsymbol{V}_1$ and $\boldsymbol{V}_2$.

For $\boldsymbol{V}_1$, we apply the mse loss to align it with fine-grained land-cover descriptions, using the text encoder to produce the semantic feature $\boldsymbol{V}_{\mathrm{LLM}}$ and compute the context mapping loss $\mathcal{L}_{\mathrm{map}}$

$$\mathcal{L}_{\mathrm{map}} = \frac{1}{d} \sum_{i=1}^{d} \|\boldsymbol{V}_1 - \boldsymbol{V}_{\mathrm{LLM}}\|_2^2 \tag{12}$$

where d represents the number of texts generated by the LLM. We perform a copy operation on $\boldsymbol{V}_1$ to match the dimensions of $\boldsymbol{V}_{\mathrm{LLM}}$. Through the constraint of $\mathcal{L}_{\mathrm{map}}$, the learnable prompts gradually utilize the internal knowledge of CLIP to generate rich contextual features consistent with LLM descriptions during optimization, thereby enabling generalization to unseen domains and classes.

For $\boldsymbol{V}_2$, we align the spectral attribute representation $\tilde{\boldsymbol{V}}$ based on L1 loss, so that when $\boldsymbol{P}_2$ encounters a new category, it can adaptively obtain the spectral information of the corresponding category from the cache according to the category semantics. The spectral attribute constraint $\mathcal{L}_{\mathrm{reg}}$ is computed as follows:

$$\mathcal{L}_{\mathrm{reg}} = \left|\boldsymbol{V}_2 - \tilde{\boldsymbol{V}}\right|. \tag{13}$$

On the image side, following the original CLIP framework, we employ a contrastive loss $\mathcal{L}_{\mathrm{cls}}$ to supervise the alignment between the output feature $\boldsymbol{x}_i$ of each image patch $\widetilde{\boldsymbol{X}}_i$ and the corresponding text feature $\boldsymbol{V}_F^c$, as shown in (14), where τ is the temperature parameter and $cos(\cdot, \cdot)$ denotes the cosine similarity. Note that the textual embedding space is treated as a domain-invariant space, and the purpose of $\boldsymbol{P}_1$ and $\boldsymbol{P}_2$ is to learn category-independent, generic spatial–spectral semantic representations. Therefore, the gradient of the contrastive loss does not propagate to the textual side, as shown in Fig. 2

$$\mathcal{L}_{\mathrm{cls}} = -\log \frac{\exp\left(\cos\left(\boldsymbol{x}_i, \boldsymbol{V}_F^c\right) / \tau\right)}{\sum_{j=1}^{C} \exp\left(\cos\left(\boldsymbol{x}_i, \boldsymbol{V}_F^j\right) / \tau\right)}. \tag{14}$$

Finally, the total loss $\mathcal{L}$ during training is the weighted sum of the above three losses, as shown in the following equation:

$$\mathcal{L} = \mathcal{L}_{\mathrm{cls}} + \lambda_1 * \mathcal{L}_{\mathrm{map}} + \lambda_2 * \mathcal{L}_{\mathrm{reg}}. \tag{15}$$

where λ_1 and λ_2 are the weights of the corresponding loss terms.

During inference, the two learned soft prompts $\boldsymbol{P}_1$ and $\boldsymbol{P}_2$, which encode domain-invariant contextual semantics from different perspectives, are concatenated with the novel category name embeddings $\boldsymbol{t}_k$ of the target domain to form enhanced textual inputs

$$\boldsymbol{T}_{1,k} = [\boldsymbol{P}_1; \boldsymbol{t}_k], \quad \boldsymbol{T}_{2,k} = [\boldsymbol{P}_2; \boldsymbol{t}_k], \quad k = 1, \ldots, K. \tag{16}$$

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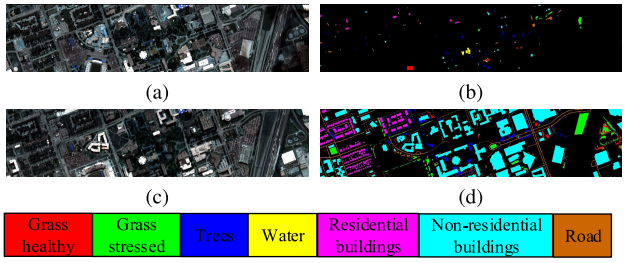


Fig. 7. Pseudo-color images and ground-truth maps of the Houston dataset. (a) Houston 2013 pseudo-color image. (b) Houston 2013 ground truth. (c) Houston 2018 pseudo-color image. (d) Houston 2018 ground truth.

TABLE I
SAMPLE DISTRIBUTION OF THE DATASET IN HOUSTON 2013 (SOURCE) AND HOUSTON 2018 (TARGET)

Class No.	Class Name	Source	Target
1	Grass healthy	345	1353
2	Grass stressed	365	4888
3	Trees	365	2766
4	Water	285	22
5	Residential buildings	319	5347
6	Non-residential buildings	408	32459
7	Road	443	6365
Total		**2530**	**53200**

These prompt-augmented sequences are passed through the text encoder to obtain two semantic embeddings $z_{1,k}^{\text{text}}$ and $z_{2,k}^{\text{text}}$, which are then fused via an element-wise summation

$$z_k^{\text{text}} = z_{1,k}^{\text{text}} + z_{2,k}^{\text{text}}. \tag{17}$$

The resulting representations z_k^{text} serve as class prototypes, which are compared with the visual feature z^{image} extracted from the image encoder. The final zero-shot classification is performed by computing their cosine similarity

$$\hat{y} = \arg\max_k \frac{\left\langle z^{\text{image}}, z_k^{\text{text}} \right\rangle}{\left\| z^{\text{image}} \right\|_2 \left\| z_k^{\text{text}} \right\|_2}. \tag{18}$$

In this way, the model leverages both the generalizable semantics encoded in the prompts and the specific class labels of unseen categories, thereby enabling zero-shot cross-scene HSIC.

IV. EXPERIMENTS

A. Datasets

1) Houston: The Houston dataset comprises HSIs acquired by different sensors over the University of Houston and its surrounding areas in 2013 [37] and 2018 [38]. The 2013 data contain 144 spectral bands with a spatial resolution of 2.5 m, while the 2018 data contain 48 spectral bands within the same wavelength range and a higher spatial resolution of 1 m. In this experiment, the 48 common spectral bands are used, focusing on an overlapping study area of 209×955 pixels. Both datasets are annotated with seven consistent land-cover classes. The number of samples per shared class is summarized in Table I. Fig. 7 presents the pseudo-color images and the corresponding ground-truth maps.

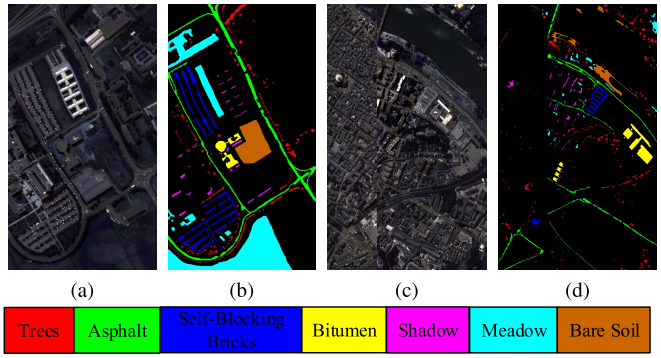


Fig. 8. Pseudo-color images and ground-truth maps of the Pavia dataset. (a) PaviaU pseudo-color image. (b) PaviaU ground truth. (c) PaviaC pseudo-color image. (d) PaviaC ground truth.

TABLE II
SAMPLE DISTRIBUTION OF THE PAVIA DATASET IN PAVIAU (SOURCE) AND PAVIAC (TARGET)

Class No.	Class Name	Source	Target
1	Tree	3064	7598
2	Asphalt	6631	9248
3	Brick	3682	2685
4	Bitumen	1330	7287
5	Shadow	947	2863
6	Meadow	18649	3090
7	Bare soil	5029	6584
Total		**39332**	**39355**

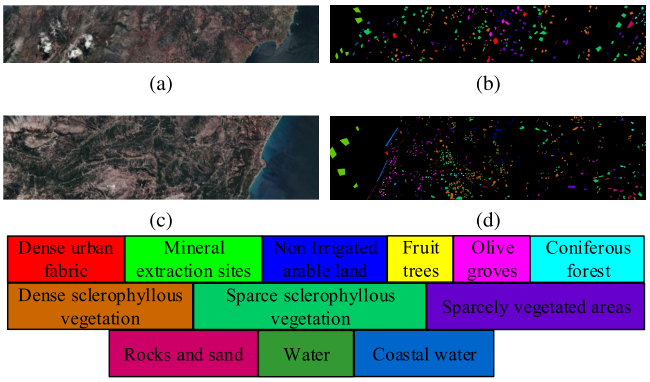


Fig. 9. Pseudo-color images and ground-truth maps of the HyRANK dataset. (a) Dioni pseudo-color image. (b) Dioni ground truth. (c) Loukia pseudo-color image. (d) Loukia ground truth.

2) Pavia: The Pavia dataset [39] consists of two scenes captured by the ROSIS sensor in northern Italy: the University of Pavia (PaviaU) and the Pavia Center (PaviaC). The PaviaU data have a spatial resolution of 1.3 m and an image size of 610×340 pixels and originally contain 103 spectral bands, while the PaviaC data have an image size of 1096×715 pixels with 102 spectral bands. To ensure consistency between the two datasets, the last spectral band of PaviaU is removed in this study. Seven consistent land-cover classes are selected. The names and sample counts of these classes are provided in Table II, and pseudo-color images along with the corresponding ground-truth maps are presented in Fig. 8.

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TABLE III
SAMPLE DISTRIBUTION OF THE HYRANK DATASET IN DIONI (SOURCE) AND LOUKIA (TARGET)

Class No.	Class Name	Source	Target
1	Dense urban fabric	1262	206
2	Mineral extraction sites	204	54
3	Non irrigated arable land	614	426
4	Fruit trees	150	79
5	Olive groves	1768	1107
6	Coniferous forest	361	422
7	Dense sclerophyllous vegetation	5035	2996
8	Sparse sclerophyllous vegetation	6374	2361
9	Sparsely vegetated areas	1754	399
10	Rocks and sand	492	453
11	Water	1612	1393
12	Coastal water	398	421
	Total	**20024**	**10317**

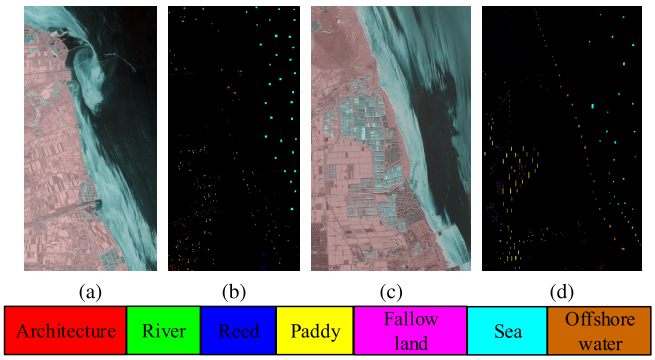


Fig. 10. Pseudo-color images and ground-truth maps of the YC dataset. (a) GFYC pseudo-color image. (b) GFYC ground truth. (c) ZYYC pseudo-color image. (d) ZYYC ground truth.

TABLE IV
SAMPLE DISTRIBUTION OF THE YC DATASET IN GFYC (SOURCE) AND ZYYC (TARGET)

Class No.	Class Name	Source	Target
1	Architecture	360	451
2	River	217	214
3	Reed	132	244
4	Paddy	832	3026
5	Fallow land	234	650
6	Sea	2395	2076
7	Offshore water	1305	1558
	Total	**39332**	**39355**

3) HyRANK: The HyRANK dataset [40], released by the ISPRS Scientific Initiatives program, is a hyperspectral remote sensing dataset acquired by the Hyperion sensor with a spatial resolution of 30 m and 176 spectral bands. Among its multiple scenes, we adopt the commonly used Dioni (250×1376) and Loukia (249×945) scenes, which share 12 land-cover classes. The sample distribution of these shared classes, along with corresponding pseudo-color images and ground-truth maps, is presented in Table III and Fig. 9, respectively.

4) YC: The YC dataset [41] comprises two subsets acquired by the AHSI onboard the GF-5 and ZY1-02D satellites. The GF-5 scene has a spatial size of 1175×585 pixels, whereas the ZY1-02D scene measures 1398×942 pixels. To establish a consistent basis for analysis, we extract the spectrally aligned portion shared by both subsets, resulting in 147 spectral bands in total, and conduct classification experiments on seven common land-cover categories. The sample distribution of these shared classes, together with the corresponding pseudo-color composites and ground-truth maps, is provided in Table IV and Fig. 10, respectively.

B. Experimental Setting

1) Base- and Novel-Class Division: In this task, following the definition of zero-shot cross-scene HSIC in Section II-A, the target domain contains novel classes that are absent from the source domain. To address this, we adopt the ZSDG strategy proposed by Maniyar et al. [42]. For each dataset, a subset of classes is designated as novel classes in the target domain, while the remaining classes are treated as base classes. During training, novel-class samples are excluded from the source domain, and the model is trained on base-class samples. Evaluation is conducted directly on the novel-class samples in the target domain. For each dataset pair, we consider multiple experimental settings, and each is defined by a different selection of novel classes.

1) *Houston:* Setting 1—(trees and residential buildings), Setting 2—(water and nonresidential buildings), Setting 3—(grass stressed and road), and Setting 4—(grass healthy and nonresidential buildings).
2) *Pavia:* Setting 1—(bricks and shadow), Setting 2—(asphalt and bricks), Setting 3—(trees and bare soil), and Setting 4—(trees and meadows).
3) *HyRANK:* Setting 1—(dense urban fabric, non irrigated arable land, coniferous forest, sparce sclerophyllous vegetation, rocks and sand, and water); Setting 2—(dense urban fabric, non irrigated arable land, olive groves, sparce sclerophyllous vegetation, rocks and sand, and water); and Setting 3—(non irrigated arable land, fruit trees, olive groves, sparce sclerophyllous vegetation, sparcely vegetated areas, and coastal water).
4) *YC:* Setting 1—(river, sea, and offshore water) and Setting 2—(architecture and reed).

2) Comparison With Sota Methods: To validate the proposed method, we compare it with four state-of-the-art DG methods for HSIs: LDGNet [10], EHSnet [11], TASA-Net [12], and ISDGS [9], as well as three state-of-the-art PL methods in natural images: CoOP [23], PromptSRC [25], and MMRL [43] to demonstrate its effectiveness.

1) *LDGNet:* It introduces a language-aware DG network with an explicit classifier, leveraging multigranularity text to improve cross-scene HSIC through visual-linguistic alignment in a unified semantic space.
2) *EHSnet:* It leverages image- and patch-level textual semantics to extract domain-invariant features through supervised contrastive learning, followed by an explicit classifier for prediction.
3) *TASA-Net:* It proposes a text-driven adaptive semantic alignment network for efficient DA and feature alignment in cross-scene HSIC by dynamically adjusting multilevel semantics.

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4) *ISDGS:* It employs a style-content randomization generator and spatial shuffling discriminator to augment domain data while preserving category semantics, leverages invariant features and a dual-sampling contrastive strategy, is followed by an explicit classifier.
5) *CoOP:* It is a context-optimization method that automates prompt engineering for VLFMs, enhancing transferability and robustness to domain shifts through data-efficient and task-specific PL.
6) *PromptSRC:* It is a self-regularizing PL framework that reduces overfitting during transfer by incorporating mutual information consistency, self-integration, and textual diversity into VLFMs like CLIP, enhancing downstream performance while preserving generalization.
7) *MMRL:* It is a multimodal representation learning framework that improves the transferability and generalization of VLFMs by introducing a shared, learnable, modality-independent space within the high-level encoder.

For all methods, we use the optimal parameter settings reported in their original articles, and when no dataset-specific parameters are provided, we perform independent parameter tuning.

3) Evaluation Metrics: To verify that the proposed method maintains robust performance on base classes while generalizing effectively to novel classes, we adopt an evaluation setup similar to the Base2Novel task [24]. Specifically, the model is evaluated separately on the base and novel classes of the target domain using overall accuracy (OA), average accuracy (AA), and Kappa coefficient (Kappa). Additionally, we compute the harmonic mean of the metrics between base and novel classes to assess the model's ability to balance both. The harmonic mean is calculated as follows, where i denotes the evaluation metric (OA, AA, or Kappa), and Base_{OA} represents its value on the base classes

$$\text{H}_i = \frac{2 \times \text{Base}_i \times \text{Novel}_i}{\text{Base}_i + \text{Novel}_i}. \tag{19}$$

4) Implementation Details: FTSA is implemented on the PyTorch platform, and all experiments in this chapter are performed on a single NVIDIA GeForce RTX 3090 24-GB GPU and a single Intel[1] Xeon[1] Silver 4210R CPU. Optimization is performed using the Adam optimizer with an initial learning rate of 0.0001 and weight decay of 0.01. The HSI patch size is 13×13, the training batch size is 128 for each of the three datasets, and the number of training epochs is 200. We adopt CLIP-ViT-B/32 as the frozen pretrained VLFM, with the spectral uniformity dimension $c = 100$. The Top-k value is set to 5 in both the SPG and SALC modules. For λ_1, we set the weights to 2.0 for Houston and YC, 2.5 for Pavia, and 1.5 for HyRANK. For λ_2, we use a unified setting of 0.5 across all datasets.

C. Experimental Results and Analysis

1) Quantitative Results: Tables V–VIII report the OA, Kappa, and AA metrics for base and novel classes, as well as their harmonic mean, across all settings on the Houston, Pavia, HyRANK, and YC datasets. The average results are also computed to comprehensively assess overall model performance on each dataset. From the experimental results, FTSA achieves the best overall performance in OA, Kappa, and AA across all four datasets, demonstrating strong advantages in ZSDG. As shown in the tables, while traditional DG methods can adapt to spatial distribution shifts and achieve good base-class accuracy, they struggle to recognize novel classes and fail to generalize in the target domain. In contrast, image–text matching-based approaches, such as FTSA, show stronger generalization. Across all four datasets, FTSA improves the harmonic mean of OA, Kappa, and AA by 2.06%–8.59%, 1.19%–20.52%, and 2.99%–9.44%, respectively, compared to the second-best model, indicating that generalized information of fine-grained texts and caches can provide domain-invariant semantic supervision. This supervision guides the soft prompts to learn generalized contextual spatial–spectral features, enabling effective novel-class recognition in the target domain.

Compared with TASA-Net, which uses fixed text templates, FTSA achieves notable improvements in both base- and novel-class performance, particularly in the Kappa and AA metrics. Specifically, FTSA improves the harmonic mean Kappa by 15.74%, 20.52%, 14.34%, and 8.69% and the harmonic mean AA by 2.99%, 9.44%, 8.64%, and 7.24% on the four datasets, respectively. These results suggest that fine-grained descriptions generated by LLMs provide richer supervision than fixed templates, enabling more balanced classification even in cases of high semantic similarity or severe class imbalance. Moreover, unlike LDGNet, which requires manual design of fine-grained descriptions, FTSA only needs a few prompt templates to automatically generate diverse linguistic representations via LLM, significantly reducing annotation costs.

Unlike FTSA, CoOP, PromptSRC, and MMRL are originally designed for RGB images and lack the capacity to capture spectral characteristics, resulting in degraded performance on HSI tasks. Moreover, these methods exhibit substantial performance fluctuations across different novel-class settings. For instance, CoOP achieves 91.95% Kappa for novel classes in setting 3 on the Pavia dataset but completely fails to recognize novel classes in settings 2 and 4. Similar instability occurs with PromptSRC on the Houston dataset (settings 1 and 2) and MMRL on the HyRANK dataset (settings 1 and 2). In contrast, FTSA learns generic contextual features via prompt tuning, demonstrating greater robustness and consistency across varying novel-class distributions.

We further evaluate all methods on the YC dataset under two challenging novel-class configurations. Setting 1 includes three water-related categories that are semantically distant from the land-cover base classes yet highly similar to each other. Most methods show limited transferability in this scenario, whereas FTSA achieves a harmonic mean OA of 87.68%, substantially outperforming all competing methods. Setting 2 involves two semantically distant classes, "Architecture" (man-made) and "Reed" (natural wetland ecosystem), both significantly different from the base

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TABLE V
QUANTITATIVE CLASSIFICATION RESULTS USING HOUSTON 2013 AS THE SOURCE DOMAIN AND HOUSTON 2018 AS THE TARGET DOMAIN

Method	Setting	Base			Novel			Harmonic		
		OA	Kappa	AA	OA	Kappa	AA	OA	Kappa	AA
EHSnet	Setting 1	81.50	61.82	77.63	0.00	0.00	0.00	0.00	0.00	0.00
	Setting 2	72.40	63.72	68.42	0.00	0.00	0.00	0.00	0.00	0.00
	Setting 3	88.30	69.25	82.51	0.00	0.00	0.00	0.00	0.00	0.00
	Setting 4	80.57	73.45	83.21	0.00	0.00	0.00	0.00	0.00	0.00
	Avg	80.69	67.06	77.94	0.00	0.00	0.00	0.00	0.00	0.00
ISDGS	Setting 1	84.51	66.50	84.05	0.00	0.00	0.00	0.00	0.00	0.00
	Setting 2	82.07	76.58	80.75	0.00	0.00	0.00	0.00	0.00	0.00
	Setting 3	88.77	71.04	86.04	0.00	0.00	0.00	0.00	0.00	0.00
	Setting 4	78.09	70.08	81.58	0.00	0.00	0.00	0.00	0.00	0.00
	Avg	83.36	71.05	**83.11**	0.00	0.00	0.00	0.00	0.00	0.00
LDGnet	Setting 1	90.11	77.19	77.20	0.00	0.00	0.00	0.00	0.00	0.00
	Setting 2	79.04	72.27	76.58	0.00	0.00	0.00	0.00	0.00	0.00
	Setting 3	89.79	73.65	84.16	0.00	0.00	0.00	0.00	0.00	0.00
	Setting 4	81.24	74.36	83.92	0.00	0.00	0.00	0.00	0.00	0.00
	Avg	**85.05**	**74.37**	80.47	0.00	0.00	0.00	0.00	0.00	0.00
TASA_net	Setting 1	76.57	42.46	46.45	81.36	52.97	73.36	78.89	47.14	56.88
	Setting 2	75.08	66.87	66.28	91.95	1.15	84.62	82.66	2.26	74.33
	Setting 3	71.27	29.09	45.86	54.58	17.05	59.50	61.82	21.50	51.80
	Setting 4	80.60	73.38	79.46	97.39	69.44	88.58	88.20	71.36	83.77
	Avg	75.88	52.95	59.51	81.32	35.15	76.52	77.89	35.57	66.70
CoOP	Setting 1	77.92	49.17	53.44	73.15	29.77	62.66	75.46	37.09	57.69
	Setting 2	54.48	37.68	42.86	78.32	0.20	66.44	64.26	0.41	52.11
	Setting 3	81.64	39.85	38.12	84.97	69.17	84.31	83.27	50.57	52.50
	Setting 4	53.84	34.12	58.72	82.53	15.41	70.71	65.17	21.23	58.72
	Avg	68.97	40.21	48.29	79.74	28.64	71.03	72.04	27.33	55.26
PromptSRC	Setting 1	75.82	29.76	45.45	78.65	48.95	72.77	77.21	37.02	55.95
	Setting 2	59.04	44.94	50.66	30.77	0.01	51.73	40.45	0.01	51.19
	Setting 3	80.28	49.00	59.64	88.45	75.85	86.91	84.16	59.53	70.74
	Setting 4	74.23	64.37	77.87	92.58	48.25	95.04	82.40	55.15	85.60
	Avg	72.34	47.02	58.41	72.61	43.27	76.61	71.06	37.93	65.87
MMRL	Setting 1	84.45	64.04	69.82	74.81	32.11	63.26	79.34	42.78	66.38
	Setting 2	68.08	57.12	57.35	11.88	−0.01	46.82	20.23	−0.02	51.56
	Setting 3	86.14	63.06	73.75	57.90	3.46	51.53	69.25	6.56	60.67
	Setting 4	74.28	64.60	78.55	96.00	0.00	50.00	83.75	0.00	61.10
	Avg	78.24	62.21	69.87	60.15	8.89	52.90	63.14	12.33	59.93
Ours	Setting 1	87.08	70.73	78.17	90.46	77.67	86.92	88.74	74.04	82.31
	Setting 2	79.44	72.76	75.24	98.65	2.11	60.72	88.01	4.09	67.20
	Setting 3	80.21	43.45	44.31	84.94	68.96	84.05	82.51	53.31	58.02
	Setting 4	77.34	68.89	59.86	98.50	79.49	87.88	86.64	73.81	71.21
	Avg	81.02	63.96	64.40	**93.14**	**57.06**	**79.89**	**86.48**	**51.31**	**69.69**

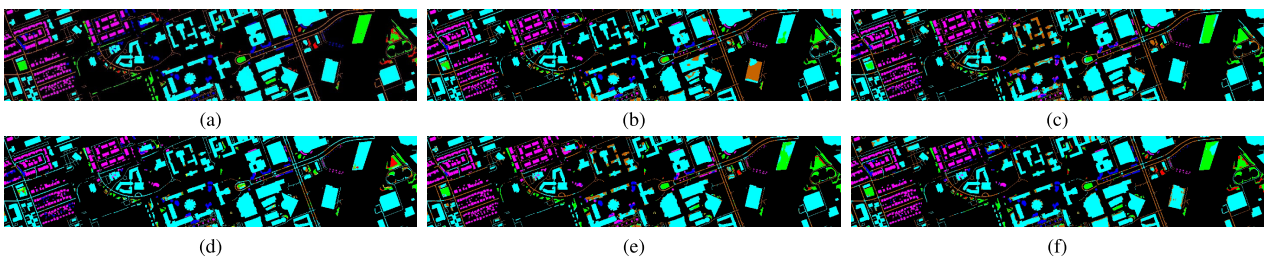


Fig. 11. Classification maps of different models in the target domain Houston 2018. (a) Ground truth. (b) TASA_net. (c) CoOP. (d) PromptSRC. (e) MMRL. (f) FTSA.

classes. FTSA achieves the highest harmonic mean OA of 87.76%, demonstrating its ability to effectively distinguish novel classes that are semantically far from base classes. Overall, the results on both settings demonstrate the robustness of FTSA under semantically distant and fine-grained novel-class conditions.

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TABLE VI
QUANTITATIVE CLASSIFICATION RESULTS USING PAVIAU AS THE SOURCE DOMAIN AND PAVIAC AS THE TARGET DOMAIN

Method	Setting	Base			Novel			Harmonic		
		OA	Kappa	AA	OA	Kappa	AA	OA	Kappa	AA
EHSnet	Setting 1	81.31	76.17	80.88	0.00	0.00	0.00	0.00	0.00	0.00
	Setting 2	90.70	87.90	87.99	0.00	0.00	0.00	0.00	0.00	0.00
	Setting 3	88.04	84.10	91.37	0.00	0.00	0.00	0.00	0.00	0.00
	Setting 4	88.84	85.38	89.95	0.00	0.00	0.00	0.00	0.00	0.00
	Avg	87.22	83.39	87.55	0.00	0.00	0.00	0.00	0.00	0.00
ISDGS	Setting 1	89.40	86.34	85.62	0.00	0.00	0.00	0.00	0.00	0.00
	Setting 2	92.85	90.68	90.20	0.00	0.00	0.00	0.00	0.00	0.00
	Setting 3	89.67	86.12	90.61	0.00	0.00	0.00	0.00	0.00	0.00
	Setting 4	90.16	87.15	91.49	0.00	0.00	0.00	0.00	0.00	0.00
	Avg	**90.52**	**87.57**	**89.48**	0.00	0.00	0.00	0.00	0.00	0.00
LDGnet	Setting 1	82.72	77.96	81.53	0.00	0.00	0.00	0.00	0.00	0.00
	Setting 2	91.12	88.42	87.60	0.00	0.00	0.00	0.00	0.00	0.00
	Setting 3	86.12	81.29	87.35	0.00	0.00	0.00	0.00	0.00	0.00
	Setting 4	82.49	77.10	81.97	0.00	0.00	0.00	0.00	0.00	0.00
	Avg	85.61	81.19	84.61	0.00	0.00	0.00	0.00	0.00	0.00
TASA_net	Setting 1	78.40	72.35	77.51	97.03	94.04	96.93	86.72	81.78	86.14
	Setting 2	88.17	84.77	87.51	90.12	67.48	79.39	89.14	75.15	83.26
	Setting 3	72.26	64.33	79.57	77.69	53.68	75.99	74.87	58.52	77.74
	Setting 4	81.33	75.64	80.53	74.52	17.79	56.71	77.78	28.81	66.55
	Avg	80.04	74.27	81.28	84.84	58.25	77.26	82.13	61.07	78.42
CoOP	Setting 1	80.41	74.78	78.44	94.70	89.42	94.85	86.97	81.45	85.86
	Setting 2	82.67	77.34	76.80	22.50	0.00	50.00	35.37	0.00	60.57
	Setting 3	69.04	56.55	58.38	95.97	91.95	96.21	80.31	70.03	72.67
	Setting 4	43.34	18.61	29.87	71.09	0.00	50.00	53.85	0.00	37.40
	Avg	68.87	56.82	60.87	71.07	45.34	72.77	64.13	37.87	64.13
PromptSRC	Setting 1	84.92	80.66	81.83	48.40	0.00	50.00	61.65	0.00	62.07
	Setting 2	84.00	78.91	77.06	76.55	−1.81	49.40	80.10	−3.71	60.21
	Setting 3	82.11	75.51	79.18	46.45	0.04	50.02	59.33	0.07	61.31
	Setting 4	80.97	74.71	74.22	71.09	0.00	50.00	75.71	0.00	59.75
	Avg	83.00	77.45	78.07	60.62	−0.44	49.86	69.20	−0.91	60.84
MMRL	Setting 1	82.70	77.67	79.80	75.32	51.31	76.03	78.84	61.80	77.87
	Setting 2	86.77	82.66	81.24	64.91	−7.26	46.60	74.27	−15.91	59.22
	Setting 3	82.41	75.96	78.31	54.10	13.45	57.17	65.32	22.85	66.09
	Setting 4	72.01	63.34	69.72	27.17	−5.57	45.33	39.45	−12.20	54.94
	Avg	80.97	74.91	77.27	55.38	12.98	56.28	64.47	14.14	64.53
Ours	Setting 1	90.01	87.21	88.53	98.77	97.54	98.74	94.19	92.09	93.36
	Setting 2	93.39	91.41	90.96	92.40	78.03	88.70	92.89	84.19	89.82
	Setting 3	77.42	70.18	73.15	97.39	94.77	97.56	86.26	80.65	83.61
	Setting 4	83.53	78.42	84.05	82.42	62.26	85.27	82.97	69.41	84.66
	Avg	86.09	81.81	84.17	**92.75**	**83.15**	**92.58**	**89.08**	**81.59**	**87.86**

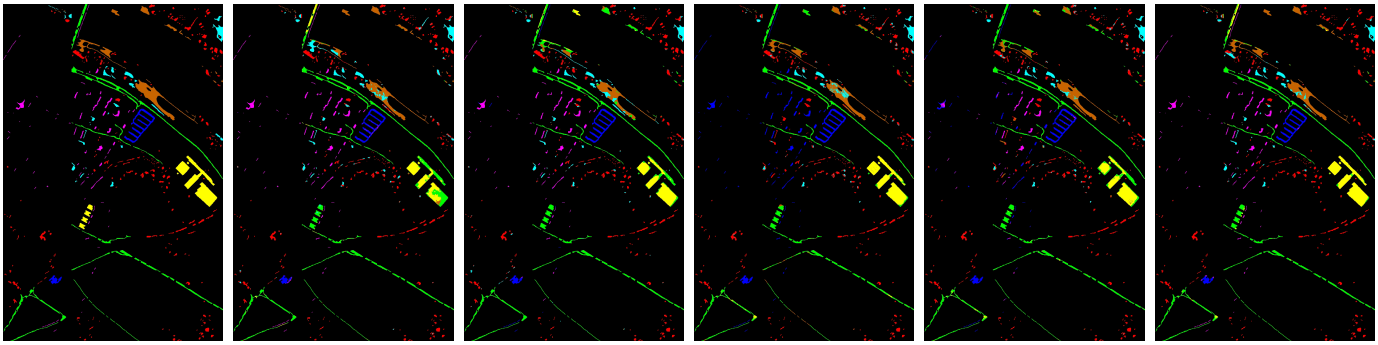
2) Visual Classification Maps: Figs. 11–14 show classification visualizations of five zero-shot methods under setting 1 across the four datasets. The results demonstrate that FTSA achieves more balanced classification between base and novel classes across all four datasets, delivering superior overall performance. Specifically, FTSA better differentiates semantically similar features, such as healthy versus stressed grass in Houston, asphalt versus bitumen in Pavia, and dense versus sparse sclerophyllous vegetation. Moreover, FTSA outperforms other methods in recognizing novel classes, which often misclassify them as majority classes—for instance, mislabeling trees as residential buildings in Houston.

Although FTSA achieves the best overall performance on all four datasets, some limitations remain. First, recognizing novel-class samples can slightly reduce the discrimination of base classes, causing FTSA's average base-class accuracy to be marginally lower than that of DG methods—an issue common to other zero-shot approaches. Additionally, FTSA struggles with classes that have very few samples; for example, on the HyRANK dataset, the scarce fruit tree samples are often

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TABLE VII
QUANTITATIVE CLASSIFICATION RESULTS USING DIONI AS THE SOURCE DOMAIN AND LOUKIA AS THE TARGET DOMAIN

Method	Setting	Base			Novel			Harmonic		
		OA	Kappa	AA	OA	Kappa	AA	OA	Kappa	AA
EHSnet	Setting 1	77.37	57.64	51.90	0.00	0.00	0.00	0.00	0.00	0.00
	Setting 2	89.99	83.25	61.90	0.00	0.00	0.00	0.00	0.00	0.00
	Setting 3	87.44	73.03	57.48	0.00	0.00	0.00	0.00	0.00	0.00
	Avg	84.93	71.31	57.09	0.00	0.00	0.00	0.00	0.00	0.00
ISDGS	Setting 1	84.08	71.42	65.22	0.00	0.00	0.00	0.00	0.00	0.00
	Setting 2	89.03	81.84	63.08	0.00	0.00	0.00	0.00	0.00	0.00
	Setting 3	87.30	72.27	53.70	0.00	0.00	0.00	0.00	0.00	0.00
	Avg	**86.80**	**75.18**	60.67	0.00	0.00	0.00	0.00	0.00	0.00
LDGnet	Setting 1	74.43	53.82	74.09	0.00	0.00	0.00	0.00	0.00	0.00
	Setting 2	88.60	81.29	62.68	0.00	0.00	0.00	0.00	0.00	0.00
	Setting 3	86.66	71.24	55.42	0.00	0.00	0.00	0.00	0.00	0.00
	Avg	83.21	68.78	**64.06**	0.00	0.00	0.00	0.00	0.00	0.00
TASA_net	Setting 1	74.49	48.39	49.98	65.86	46.74	34.69	69.91	47.55	40.96
	Setting 2	80.52	64.69	36.21	58.17	23.28	33.71	67.54	34.24	34.92
	Setting 3	79.89	48.28	36.47	39.71	7.82	16.67	53.05	13.45	22.88
	Avg	78.30	53.79	43.03	54.58	25.95	28.36	63.50	31.75	32.92
CoOP	Setting 1	72.54	44.29	45.82	52.50	28.07	25.99	60.91	34.36	33.17
	Setting 2	79.04	64.09	36.43	54.02	21.18	32.60	64.17	31.83	34.41
	Setting 3	83.37	61.94	46.36	64.08	47.18	43.95	72.46	53.56	45.12
	Avg	78.31	56.77	42.87	56.87	32.14	34.18	65.85	39.92	37.57
PromptSRC	Setting 1	73.95	46.82	46.80	56.15	39.86	32.97	63.83	43.06	38.69
	Setting 2	79.60	64.41	33.98	58.02	28.30	41.14	67.12	39.33	37.22
	Setting 3	85.70	66.53	47.58	63.29	43.08	34.06	72.81	52.30	39.70
	Avg	79.75	59.25	42.79	59.15	37.08	36.06	67.92	44.90	38.54
MMRL	Setting 1	80.60	66.04	65.87	55.88	37.73	28.37	66.00	48.02	39.66
	Setting 2	81.12	67.38	38.68	49.26	0.00	16.67	61.30	0.00	23.30
	Setting 3	86.32	68.19	48.60	55.47	36.81	34.80	67.54	47.81	40.56
	Avg	82.68	67.20	51.05	53.54	24.85	26.61	64.95	31.94	34.51
Ours	Setting 1	73.67	46.44	47.03	76.92	63.36	44.83	75.26	53.60	45.90
	Setting 2	85.88	75.86	51.68	55.33	21.09	32.42	67.30	33.00	39.84
	Setting 3	85.06	64.98	46.85	63.13	42.89	33.33	72.48	51.67	38.95
	Avg	81.54	62.43	48.52	**65.13**	**42.45**	**36.86**	**71.68**	**46.09**	**41.56**



(a) (b) (c) (d) (e) (f)

Fig. 12. Classification maps of different models in the target domain PaviaC. (a) Ground truth. (b) TASA_net. (c) CoOP. (d) PromptSRC. (e) MMRL. (f) FTSA.

misclassified as dense sclerophyllous vegetation. This is likely due to FTSA's reliance on source-domain samples—when sample sizes are limited, the SALC module has difficulty with aggregating domain-invariant spectral attributes effectively, hindering generalization under spectral domain shifts.

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TABLE VIII
QUANTITATIVE CLASSIFICATION RESULTS USING GFYC AS THE SOURCE DOMAIN AND ZYYC AS THE TARGET DOMAIN

Method	Setting	Base			Novel			Harmonic		
		OA	Kappa	AA	OA	Kappa	AA	OA	Kappa	AA
EHSnet	Setting 1	94.70	89.50	89.72	0.00	0.00	0.00	0.00	0.00	0.00
	Setting 2	88.38	83.77	78.48	0.00	0.00	0.00	0.00	0.00	0.00
	Avg	**91.54**	**86.64**	**84.10**	0.00	0.00	0.00	0.00	0.00	0.00
ISDGS	Setting 1	95.49	90.61	89.51	0.00	0.00	0.00	0.00	0.00	0.00
	Setting 2	86.56	80.91	72.95	0.00	0.00	0.00	0.00	0.00	0.00
	Avg	91.03	85.76	81.23	0.00	0.00	0.00	0.00	0.00	0.00
LDGnet	Setting 1	85.87	79.73	71.74	0.00	0.00	0.00	0.00	0.00	0.00
	Setting 2	80.85	72.20	58.46	0.00	0.00	0.00	0.00	0.00	0.00
	Avg	83.36	75.97	65.10	0.00	0.00	0.00	0.00	0.00	0.00
TASA_net	Setting 1	76.71	38.21	42.31	89.71	80.82	83.30	82.70	51.88	56.12
	Setting 2	84.30	77.31	63.46	93.38	86.08	94.90	88.61	81.46	76.06
	Avg	80.51	57.76	52.89	91.55	83.45	**89.10**	85.66	66.67	66.09
CoOP	Setting 1	67.54	44.17	50.05	69.23	0.00	25.00	68.37	0.00	33.34
	Setting 2	80.06	70.84	60.41	35.68	0.62	50.44	49.37	1.24	54.98
	Avg	73.80	57.51	55.23	52.46	0.31	37.72	58.87	0.62	44.16
PromptSRC	Setting 1	89.66	80.16	63.71	71.40	10.78	30.27	79.50	19.00	41.04
	Setting 2	82.32	75.51	73.28	38.42	3.64	52.55	52.39	6.94	61.21
	Avg	85.99	77.84	68.50	54.91	7.21	41.41	65.95	12.97	51.13
MMRL	Setting 1	81.93	58.69	48.38	53.95	0.00	33.33	65.06	0.00	39.47
	Setting 2	81.17	72.84	72.61	35.11	0.00	50.00	49.07	0.00	59.22
	Avg	81.55	65.77	60.50	44.53	0.00	41.67	57.07	0.00	49.35
Ours	Setting 1	81.97	60.25	63.06	94.23	88.95	77.65	87.68	71.84	69.60
	Setting 2	84.54	77.83	68.34	91.22	79.93	88.35	87.76	78.87	77.06
	Avg	83.26	69.04	65.70	**92.73**	**84.44**	83.00	**87.72**	**75.36**	**73.33**

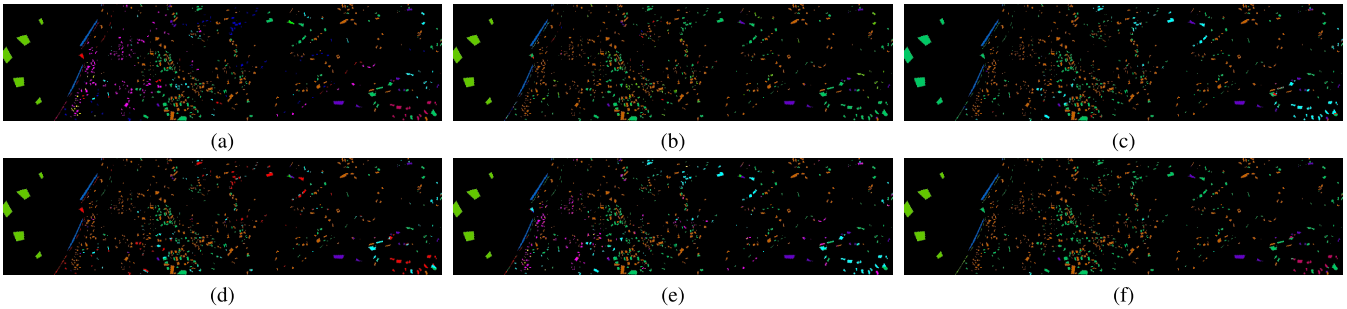
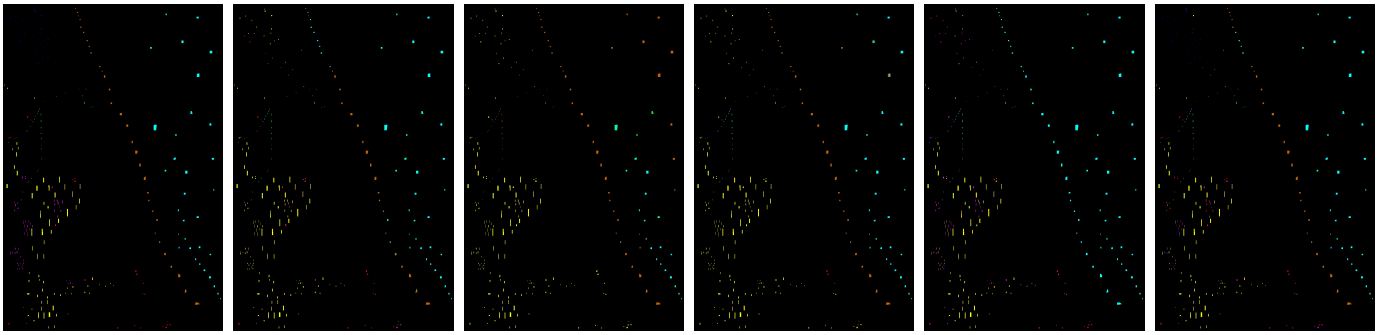


Fig. 13. Classification maps of different models in the target domain Loukia. (a) Ground truth. (b) TASA_net. (c) CoOP. (d) PromptSRC. (e) MMRL. (f) FTSA.

3) Complexity Analysis: To assess the computational overhead of the proposed FTSA, we conducted a quantitative evaluation of its parametric efficiency, training efficiency, and inference throughput on the Pavia dataset. Table IX summarizes the number of trainable parameters, average epoch duration, floating-point operations (FLOPs), training memory footprint, and inference throughput [frames per second (FPS)] for each comparative method. In terms of parameter efficiency, EHSnet, ISDGS, LDGNet, and TASA-Net are DG methods that train all model parameters. EHSnet, LDGNet, and TASA-Net partially adopt pretrained CLIP layers (e.g., the first three) and perform full fine-tuning, resulting in increased parameter counts, while ISDGS uses lightweight modules under an adversarial scheme. In contrast, CoOP, PromptSRC, MMRL, and FTSA freeze the CLIP backbone and update only prompt modules, significantly reducing trainable parameters.

In terms of training efficiency, EHSnet, LDGNet, and TASA-Net exhibit higher memory consumption due to full CLIP fine-tuning. CoOP, PromptSRC, MMRL, and FTSA incur greater FLOPs because they load the full 12-layer CLIP backbone. PromptSRC has longer epoch times due to its smaller batch sizes, while MMRL introduces higher computational overhead by utilizing both frozen and tuned CLIP encoders during training. Compared with other prompt-based methods, FTSA achieves the best classification performance

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(a) (b) (c) (d) (e) (f)

Fig. 14. Classification maps of different models in the target domain ZYYC. (a) Ground truth. (b) TASA_net. (c) CoOP. (d) PromptSRC. (e) MMRL. (f) FTSA.

TABLE IX
COMPARISON OF MODEL COMPLEXITY AMONG DIFFERENT METHODS ON THE PAVIA DATASET

Method	Tunable Params (M)	Memory (MB)	FLOPs	Train Time (s/epoch)	FPS (128 BS)
EHSnet	36.14	8941	206.12M	139.66	1196.01
ISDGS	0.68	2065	68.73M	26.90	9858.18
LDGNet	36.81	11197	89.42M	137.39	2377.64
TASA-Net	36.79	5733	1.97G	176.38	2611.80
CoOP	0.01	1951	6.82G	140.99	1748.59
PromptSRC	0.02	2751	15.37G	1196.35	933.40
MMRL	4.99	3543	110.24G	361.51	956.62
Ours	0.96	5331	11.06G	166.61	433.06

with less training time and moderate memory usage by employing larger batch sizes.

Beyond training efficiency, we also evaluated the inference efficiency of FTSA. Inference speed was quantified in FPS with a standardized batch size of 128. FTSA achieves 433.06 frames/s, slower than several state-of-the-art methods. This is mainly due to the dynamic k-NN graph construction in the SPG module, which introduces additional computational overhead during inference. While not the fastest, our method delivers superior classification accuracy, reflecting a tradeoff between inference speed and the enhanced feature quality gained from fine-grained text and spectral attribute alignment. More efficient band clustering strategies could potentially alleviate this limitation in future work.

To assess the computational cost of integrating LLMs into FTSA, we measure the time and token consumption for generating fine-grained text descriptions on the Pavia dataset. On average, each description takes 0.72 s and consumes 36.5 tokens. Notably, our method leverages an LLM API to generate these descriptions, significantly reducing the cost compared to manually designing complex descriptions for each category.

V. DISCUSSION

To further evaluate the proposed model, this section presents a series of experiments focusing on the effectiveness of individual modules, architectural design, parameter sensitivity, and visualization analysis.

A. Ablation Study

To evaluate the contribution of key modules in the proposed FTSA for zero-shot cross-scene HSIC, we conduct ablation experiments. The SPG module, fine-grained text supervision introduced by LLM, and the SALC module are identified as core components. Accordingly, five model variants with different module combinations are designed for evaluation, as detailed next.

1) M1: Remove all modules, use only the original frozen CLIP for classification, select the RGB bands for the HSI patch and interpolate to 224×224 to fit its input dimension, and use a fixed template of "an HSI of {class}" on the text side.

2) M2: Based on M1, the SPG module is added to the image encoder.

3) M3: Based on M2, soft prompt P_1 is introduced in the text encoder, fine-grained texts are introduced as its supervisory signals, and the input is replaced with a fixed template in which a soft prompt splices the category names (as described in Section III-E).

4) M4: Based on M2, soft prompt P_2 is introduced in the text encoder, the spectral attribute information cached in the SALC is used as its supervisory signal, and the text input is replaced with a soft prompt concatenated with the category name.

5) M5: As in the proposed method, all modules are included.

The harmonic mean OA, Kappa, and AA metrics for each model under setting 1 across the Houston, Pavia, and HyRANK datasets are presented in Table X. The results demonstrate that each module contributes to performance gains, leading to improved classification accuracy. The results of M1 indicate that the frozen CLIP model, despite its strong zero-shot performance on natural images, performs poorly on HSIs due to its limited capacity to model high-dimensional spectral features. In HSIC tasks, subtle spectral differences are often crucial for distinguishing land-cover classes. M2 enhances CLIP via the SPG module, converting spectral vectors into prompt tokens that are fused with patch tokens and fed into the encoder, which leads to improved performance over M1. M3 replaces fixed prompts with learnable soft prompts and introduces fine-grained texts as supervision, achieving better performance than fixed templates. The M4

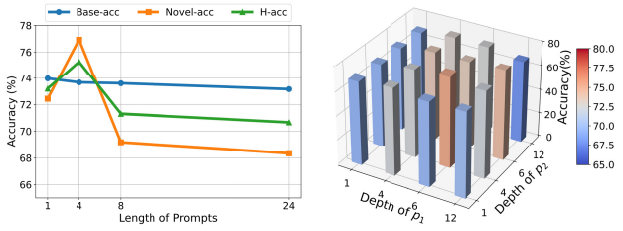
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TABLE X
ABLATION STUDY OF THREE MODULES AND SOFT PROMPT ACROSS THREE DATASETS

Model	SPG	LLM	SALC	Soft Prompt	Houston			Pavia			HyRANK		
					OA	Kappa	AA	OA	Kappa	AA	OA	Kappa	AA
M1	✗	✗	✗	✗	21.40	0.00	27.76	28.64	1.92	26.57	2.10	-0.69	14.97
M2	✓	✗	✗	✗	81.74	58.10	71.42	80.28	62.18	80.17	67.68	45.95	39.74
M3	✓	✓	✗	✓	87.79	71.04	80.83	92.26	88.49	91.42	72.70	49.91	40.48
M4	✓	✗	✓	✓	84.68	62.12	73.43	87.42	82.39	87.02	72.55	50.93	40.40
M5	✓	✓	✓	✓	**88.74**	**74.04**	**82.31**	**94.19**	**92.09**	**93.36**	**75.26**	**53.60**	**45.90**

TABLE XI
PERFORMANCE COMPARISON OF LLMS ON VARIOUS DATASETS WITH HARMONIC MEAN OA FOR SETTING 1

LLM	Houston	Pavia	HyRANK	YC
LLaMA-3	88.44	93.58	72.32	88.29
Qwen-3	86.72	**94.26**	72.42	**89.15**
Deepseek-V3	**88.74**	94.19	**75.26**	87.68



(a) (b)

Fig. 15. Effect of prompt length and depth on classification accuracy on the HyRANK dataset. (a) Accuracy on base and novel classes with varying prompt lengths. (b) Harmonic mean OA across different prompt depths.

model also adopts soft prompts and aligns them with spectral attribute features in the SALC module.

To further analyze the collaborative effect between SPG and SALC, we compare M2 (SPG only) with M4 (SPG + SALC). The results show that introducing the SALC module consistently improves performance, with the Houston OA increasing from 81.74 to 84.68. This demonstrates that SALC effectively leverages the spectral tokens generated by SPG to refine spectral attribute representations, leading to enhanced classification performance. Furthermore, comparing M3 (SPG + LLM) with M5 (Full Model) shows that SALC provides an additional performance gain (+0.95 OA on Houston) even under strong LLM supervision. This indicates that SPG and SALC are not redundant but instead form a complementary and synergistic mechanism, where SPG provides high-quality spectral tokens and SALC utilizes them to regularize and guide text PL.

Despite these improvements, M3 slightly outperforms M4 in some cases, likely because fine-grained textual descriptions provide richer supervision than the spectral cache. In addition, textual descriptions are more decoupled from image features, reducing domain-specific biases and spurious correlations in the supervision. Ultimately, M5, the proposed FTSA, achieves the highest classification performance. This demonstrates that the textual supervision provided by the LLM and the generic spectral attributes cached by the SALC module are complementary, enabling the model to learn a more robust semantic space and adapt better to scene distribution shifts, thereby improving cross-scene HSIC efficiency and zero-shot generalization to novel categories.

B. Choice of LLM for Texts Generation

To evaluate the impact of different LLMs on the performance of FTSA, we conduct an ablation study comparing two open-source LLMs: *LLaMA3* [44] and *Qwen3* [45]. Table XI demonstrates that FTSA is insensitive to the choice of LLM, as all evaluated models generate fine-grained textual descriptions that provide effective supervision signals for the training process. Across four benchmark datasets, FTSA achieves consistent performance with minimal variation, demonstrating the robustness and general applicability of the approach for ZSDG using different pretrained LLMs.

C. Length and Depth of Parallel Prompts

Parallel soft prompts P_1 and P_2 progressively learn cross-domain and cross-category representations by aligning fine-grained texts and cached spectral attribute knowledge, respectively. Accordingly, the length and depth of the soft prompts affect their ability to capture generalizable features. To investigate this effect, we conduct repeated experiments under setting 1 on the HyRANK dataset to evaluate how different prompt lengths, along with the respective depths of P_1 and P_2, influence the model's classification performance. The experimental results are shown in Fig. 15. Fig. 15(a) illustrates the model's recognition accuracy on base and novel classes in the target domain with varying prompt lengths. As the prompt length increases, the model's accuracy on base classes remains stable, while its performance on novel classes drops significantly. Notably, accuracy on novel classes decreases by 8.57% when the prompt length increases from 4 to 24. This decline is attributed to the increased number of trainable parameters, which lead the model to overfit base-class samples during training and reduce its generalization to novel classes. Due to differences in feature distribution complexity across datasets, the optimal prompt length therefore varies.

In addition to prompt length, prompt depth also influences the tradeoff between base and novel classes, and selecting an appropriate depth is critical for model performance. Fig. 15(b) illustrates the effect of parallel prompts on harmonic mean

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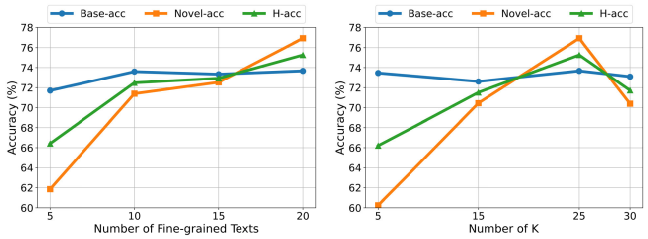


Fig. 16. (a) Effect of the number of text descriptions on classification accuracy on the HyRANK dataset. (b) Effect of the number of cached attribute entries (K) on classification accuracy on the HyRANK dataset.

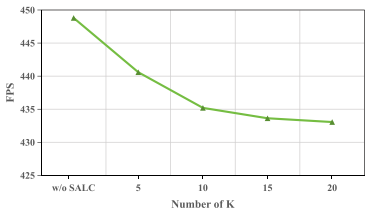


Fig. 17. Effect of cache size on inference speed (FPS) during the testing phase.

OA across different depths. The model performs poorly when either P_1 or P_2 is set to an overly shallow or deep depth (edges of the bar chart), while performance generally improves as their depths increase. Since P_1 and P_2 align fine-grained textual features and spectral attribute features, respectively, and the complexity of these modalities differs, their optimal depths vary. Prompts aligned with richer supervisory signals typically require greater depth to achieve accurate alignment. For the length of the prompts, we set 8 for Houston, Pavia, and YC, and 4 for HyRANK. The depth of P_1 is consistently set to 6 across all datasets. Regarding the depth of P_2, it is set to 8 for Houston, 4 for HyRANK, and 6 for both Pavia and YC.

D. Number of Fine-Grained Texts Provided by LLM

Due to the stochastic nature of LLMs, higher temperature yields more diverse category descriptions, enriching the supervision for soft prompts. To investigate the impact of textual diversity, we conduct repeated experiments on the HyRANK dataset (setting 1) using 5, 10, 15, and 20 fine-grained texts per base class. As shown in Fig. 16(a), increasing the number of descriptions consistently improves classification performance on both base and novel classes in the target domain—particularly for novel classes, where accuracy improves by up to 15.07%. These results indicate that richer textual supervision enhances the model's cross-domain and cross-category generalization. For the number of fine-grained texts provided by the LLM, we use 20 for Houston and HyRANK and 15 for Pavia and YC.

E. Impact of Cache Size in the SALC Module

The SALC module maintains a cache storing K spectral attribute features, gradually aggregating similar spectral visual attributes during training. The cache size K controls

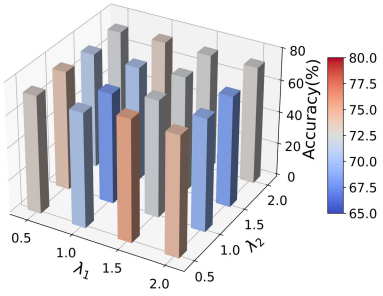


Fig. 18. Effect of varying loss term weights on classification accuracy on the HyRANK dataset.

the granularity of local spectral feature representation, with larger values introducing more attribute entries to refine local attribute representations. Ablation experiments varying the number of cached attribute terms K are presented in Fig. 16(b). Recognition performance improves as K increases, but drops beyond a certain threshold. This decline is attributed to excessive fragmentation of spectral attributes, where overly fine splits scatter semantic information, making it difficult for prompt vectors to accurately recover target semantics and thus reducing recognition accuracy. For the cache size K, we set it to 20 for Houston, Pavia, and YC datasets, and 25 for HyRANK, based on performance observations.

The choice of cache size impacts both recognition performance and inference efficiency. We evaluated the tradeoff by measuring inference speed at different cache sizes. As shown in Fig. 17, increasing the cache size causes only a slight decrease in FPS, from 448.80 without SALC to 433.06 FPS at $K = 20$. This minor reduction in inference speed is outweighed by the improvement in classification accuracy due to better local attribute representations from larger cache sizes.

F. Weights of the Loss Function Terms

λ_1 and λ_2 are the weighting terms of the total loss for the context mapping loss $\mathcal{L}_{\text{map}}$ and the spectral attribute constraint loss $\mathcal{L}_{\text{reg}}$, respectively. To assess the sensitivity of the proposed method to the loss weighting parameters, we conduct a sensitivity analysis under setting 1 on the HyRANK dataset. The values of λ_1 and λ_2 are varied in 0.5, 1.0, 1.5, 2.0. The results, shown in Fig. 18, report the harmonic mean OA achieved under different weight combinations. The experimental results show that the two loss weights have distinct but significant effects on model performance. As λ_1 increases, the model is encouraged to align more strongly with fine-grained texts, enhancing the learning of generic contextual features. However, when λ_1 becomes too large (e.g., 2.0), performance degrades, suggesting that excessive regularization interferes with the main task learning. For λ_2, a small weight improves model performance, but larger values often lead to performance fluctuations or degradation. This is likely due to overalignment with spectral cache entries in the SALC module, which causes the soft prompt P_2 to overfit the spectral features of base classes in the source domain, capturing spurious, nongeneralizable associations. As a result,

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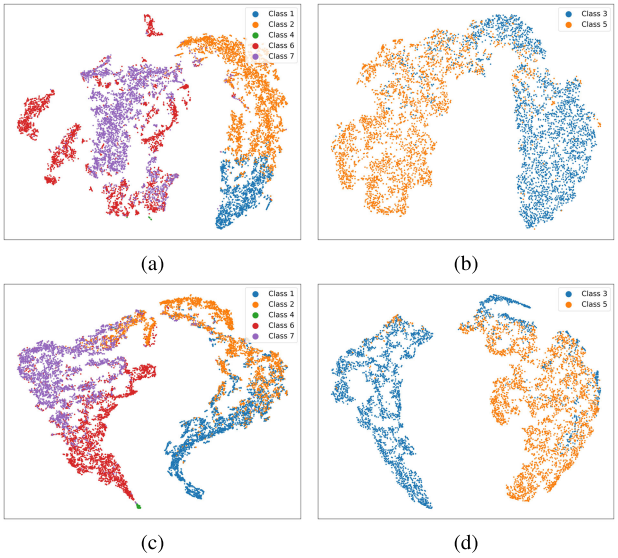


Fig. 19. T-SNE visualization of class separability on the Houston dataset. (a) and (b) Base- and novel-class distributions of the original Houston 2018 data under Setup One, respectively. (c) and (d) Feature distributions of base and novel classes after inference using the trained FTSA.

the model's ability to generalize to the target domain and novel classes is compromised.

G. Visualization Analysis

1) T-SNE Visualization: To intuitively demonstrate the category separability of the semantic space learned by FTSA, we apply t-SNE on the Houston dataset to visualize the feature distributions of base and novel classes in the target domain, before and after inference. The results are shown in Fig. 19. In the original data, some categories exhibit clustering tendencies, but considerable interclass overlap remains—particularly between spectrally similar classes such as nonresidential buildings (category 6) and roads (category 7). After applying FTSA, feature embeddings show enhanced intraclass compactness and interclass separability. Notably, novel classes unseen during training also form distinct and compact clusters, demonstrating FTSA's robustness and generalization, and highlighting its strong zero-shot transferability.

2) Cache Property Item Visualization: To evaluate whether the cached attribute entries in the SALC module capture generalizable spectral features and whether the model can selectively attend to these entries based on category semantics, we visualize the attention weights between the CLS tokens from the trained image encoder and the cached entries across all target domain samples in the Houston, Pavia, and HyRANK datasets. The weights are averaged by category, and the results are shown in Fig. 20. The results reveal distinct activation patterns of attribute terms across different categories, suggesting that the model exhibits category-aware behavior. For example, in the HyRANK dataset, categories 10 (water) and 11 (coastal water) show strong activation on the 14th attribute term, indicating that it likely encodes water-related spectral features. Similarly, categories 4–8 (fruit trees, olive groves, coniferous forest, and dense and sparse sclerophyllous vegetation) consistently attend to the 15th term, implying its association with vegetation-specific attributes. Attribute terms

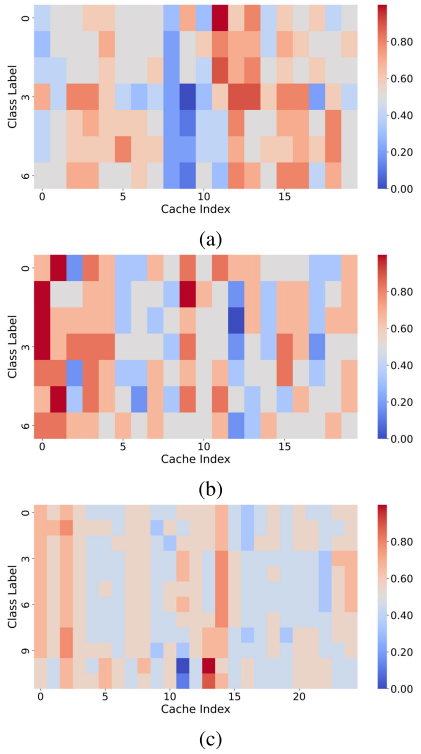


Fig. 20. Attention maps of cached attribute items for different land-cover classes across datasets.

1 and 3 receive moderate attention across multiple categories, suggesting that they capture more general or shared spectral characteristics.

However, the figures also reveal that certain attribute terms (e.g., terms 18 and 20 in HyRANK) exhibit uniformly low responses across all categories, suggesting that these terms may have failed to learn meaningful semantics. Nonetheless, most terms show category-specific activation, validating the effectiveness of the proposed local sparse matching-based attribute update strategy.

VI. Conclusion

In this study, we propose FTSA, a cross-scene HSIC method for ZSDG. Following a prompt-based fine-tuning paradigm, it introduces two parallel learnable soft prompts to extract generalizable attribute representations from fine-grained textual descriptions and visual spectral features, which adaptively generate domain-invariant category semantics for effective visual-textual alignment. FTSA demonstrates strong performance on multiple benchmarks, particularly in balancing recognition between base and novel classes and distinguishing semantically similar categories.

However, introducing LLMs increases computational overhead. Future work should explore remote sensing–specific VLFMs and more efficient fine-tuning strategies to improve both performance and efficiency in zero-shot cross-scene HSIC.

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